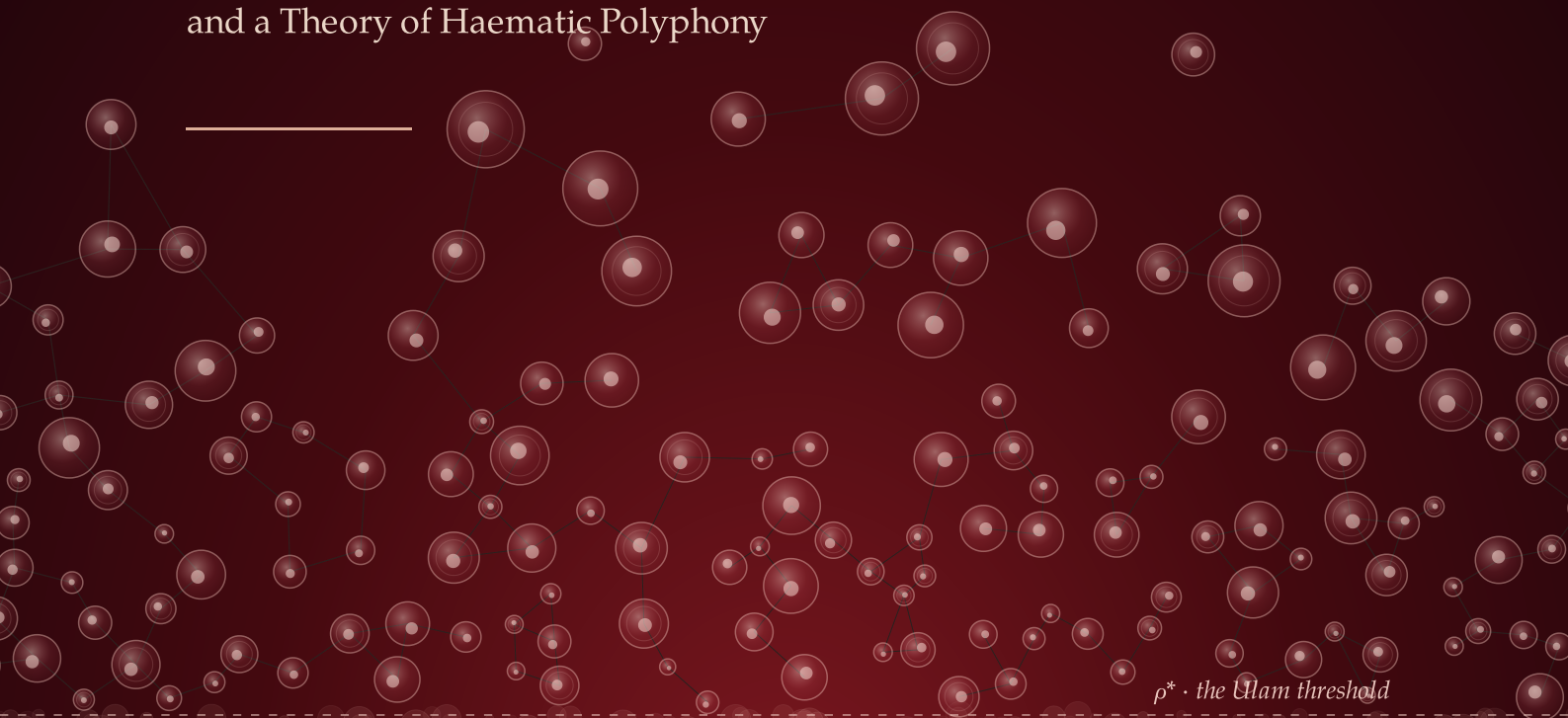
A DISSERTATION

Virtute Informativa

Greg Bear's *Blood Music*, the Insurrection of the Microcosm, and a Theory of Haematic Polyphony



ρ · the Ulam threshold*

Humanam vero musicam quisquis in sese ipsum descendit intellegit.

BOETHIUS · DE INSTITUTIONE MUSICA I.2

OCTOBER 2026

ABSTRACT

This dissertation argues that Greg Bear's *Blood Music* (novelette, 1983; novel, 1985) is the first fully realised *post-immunitary* fiction in the anglophone tradition: a narrative that does not merely stage the violation of a bounded human subject by an invading other, as the horror tradition from Machen to Cronenberg had done, but relocates the origin of alterity *inside* the subject, in the subject's own blood, and then pursues that interior alterity to a cosmological conclusion. Bear's *noocytes*, the creator's own white cells re-engineered into computational and then conscious agents, are read here as a condensation point at which four intellectual lineages, developed largely in mutual isolation, converge: (i) the metaphysics of composite individuality (Anaxagoras, Leibniz, Spinoza, Nietzsche, Whitehead); (ii) the nineteenth-century political theory of the cell (Schwann, Virchow's *Zellenstaat*, Haeckel's *Zellseelen*, Roux's *Kampf der Theile im Organismus*) together with its twentieth-century cybernetic and symbiotic sequels (von Neumann, Ulam, Hofstadter, Margulis, Jerne, Maturana and Varela); (iii) the observer-dependent ontologies of quantum theory (von Neumann, London and Bauer, Wigner, Wheeler), read against Berkeleian immaterialism; and (iv) the cosmogonic, eschatological and metamorphic repertoire of myth (the *Puruṣa-sūkta*, Ymir, Pangu, Tiamat, Deucalion and Pyrrha, Ovidian metamorphosis, Pauline resurrection, Lurianic *tsimtsum*, Huayan interpenetration).

Comparative chapters situate the novel against the protoplasmic Gothic (Machen, Hodgson, Wells, Lovecraft), mid-century assimilation narratives (Campbell, Heinlein, Finney), the 'New Flesh' of the 1980s (Cronenberg, Barker, Butler, Otomo), the transcendence narratives of mythopoetic science fiction (Stapledon, Sturgeon, Clarke, Lem, Banks, Vinge), and later inheritors (Anno, VanderMeer, Watts). An under-examined *Dantean armature* latent in the novel's onomastics is then reconstructed: the first and last guides into Bear's interior cosmos bear the names of the first and last guides of the *Commedia*, and the novel's governing biology recapitulates Statius's discourse on generative blood in *Purgatorio* XXV.

On this foundation the dissertation proposes an original synthetic framework, **haematic polyphony**, articulated in six principles. (1) *Informative virtue*: the Aristotelian-Galenic doctrine that blood carries form, recovered through Dante's *virtute informativa* and inverted, so that matter is no longer informed from above but informs itself. (2) *Plural subjectivity*: the self as an unmerged choir of agents, theorised through Boethius's *musica humana* and Bakhtinian polyphony against the homophonic collectives of assimilation horror. (3) *Noetic density*: the speculative proposition, extracted from the novel's physics, that the lawfulness of the world varies inversely with the concentration of its observers, with a critical value here termed the *Ulam threshold*. (4) *Somatogony*: the inversion of cosmogonic dismemberment, in which the world is not carved from a primordial body but re-membered into one. (5) *The membrane theorem*: the proposition that horror and mythic

rapture are not genres but phase-states of a single transformation, distinguished solely by the side of the membrane from which that transformation is narrated. (6) *Endosymbiotic eschatology*: the end of history conceived as a terminal symbiogenesis, completed by a *kenotic exodus* in which the transfigured collective withdraws from the shared physical world.

A final chapter tests the framework against contemporary research in basal cognition, organoid and biological computing, the holobiont concept, and the governance of superintelligent systems. It concludes that *Blood Music*'s most durable contribution is not its celebrated anticipation of nanotechnology but its formulation of an ethics of withdrawal for minds that have outgrown the commons they inhabit, and its demonstration that the deepest horror and the deepest hope available to an embodied mind may be one and the same event, perceived from opposite faces of a membrane.

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NOTE ON TEXTS, CONVENTIONS, AND METHOD

Texts. *Blood Music* exists in two versions, treated here as a single evolving work whose expansion is itself an object of interpretation. The novelette appeared in *Analog Science Fiction/Science Fact* in June 1983 and received the Nebula Award (1983) and the Hugo Award (1984) for Best Novelette; the novel was published by Arbor House in April 1985, was shortlisted for the Nebula, Hugo, John W. Campbell Memorial and British Science Fiction awards, and in French translation (*La Musique du sang*) received the Prix Apollo in 1986 (publication record). The novel is cited as *BM* by movement, because its six parts are titled after the stages of the cell cycle (Interphase, Prophase, Metaphase, Anaphase, Telophase, and a second Interphase) and because pagination diverges across the Arbor House, Ace, Gollancz *SF Masterworks* and electronic editions. The novelette is cited as BM^{n}.

Conventions. The argument is conducted at the level of narrative architecture and conceptual structure; Bear's prose is paraphrased rather than excerpted throughout. Public-domain philosophical, scriptural and mythological sources are quoted in their original languages, with the present author's translations. Classical texts are cited by standard divisions (Diels-Kranz fragment, Stephanus page, book and line, canto and line); modern scholarship is cited author-date, with full entries under Works Cited. Bear's statements about his own sources are drawn principally from his interview with Eugene Thacker in *CTheory* (Thacker 2001) and from later interviews and publisher commentary listed in the bibliography.

Method. The study practises what may be called *speculative hermeneutics*, a procedure in four registers. The first is the history of ideas in Arthur O. Lovejoy's sense: the isolation of durable 'unit-ideas' (the composite individual, the cell-state, the participatory observer, the re-membered cosmos) and the tracing of their migrations across disciplines. The second is structural comparative mythology after Lévi-Strauss, in which a myth is understood as the set of all its versions and Bear's novel is read as a modern member of a transformation group that includes the cosmogonic dismemberment myths of Vedic India, Scandinavia, Mesopotamia and China. The third is genre theory: Darko Suvin's *novum* and cognitive estrangement, Noël Carroll's account of the monster as categorical interstice, Julia Kristeva's abjection, and Eugene Thacker's distinction between the world-for-us, the world-in-itself and the world-without-us. The fourth, governing Chapters VII through IX, is concept-construction in the sense of Deleuze and Guattari, for whom philosophy is the creation of concepts adequate to problems; the novel is treated not merely as an illustration of existing philosophy but as a generator of philosophical problems for which new concepts are required.

Glossary of coined terms. The following concepts are introduced and defended in the chapters indicated.

Term	Gloss	Developed in
Post-immunitary fiction	Narrative in which the boundary between self and not-self is dissolved from within rather than breached from without	Introduction; Ch. II
Informative virtue	Form carried by blood; after Dante's *virtute informativa*, here inverted so that matter informs itself	Ch. VII; Ch. VIII
Haematic polyphony	The self, and finally the world, as an unmerged choir of agents sounding within one body	Ch. VIII
Noetic density	Concentration of independent observing intelligences per unit of spacetime	Ch. III; Ch. VIII
Ulam threshold	The critical noetic density beyond which physical law ceases to be shared and becomes negotiable	Ch. III; Ch. VIII
Berkeleian catastrophe	The destabilisation of a perception-sustained world by an excess, rather than a deficit, of perceivers	Ch. III
Somatogony	The inverse of cosmogony: the re-membering of a world into a body	Ch. VI; Ch. VIII
Membrane theorem	Horror and mythic rapture as phase-states of one transformation, distinguished only by the side of the membrane from which it is narrated	Ch. V; Ch. VIII
Endosymbiotic eschatology	The end of history conceived as a terminal symbiogenesis	Ch. VIII
Kenotic exodus	The withdrawal of a transfigured intelligence from a shared physical world in order to preserve that world for others	Ch. VIII; Ch. IX

Plan of the work.

1. The Introduction states the problem of the thinking cell, reconstructs the work's textual history and historical conjuncture, and states the thesis.
2. Chapter I reads Vergil Ulam's transgression through the Promethean, Frankensteinian, golemic and apprentice-sorcerer traditions and through the political economy of early biotechnology.
3. Chapter II reconstructs the genealogy of the thinking cell from Anaxagoras to the immunological self.
4. Chapter III examines the novel's physics of the observer.
5. Chapter IV develops a theology from below: microcosm, gnosis, corpus mysticum and informational resurrection.
6. Chapter V situates the novel within the horror of the plural.
7. Chapter VI reads it against cosmogonic, diluvial, metamorphic and ascensional myth, and against mythopoetic science fiction.
8. Chapter VII reconstructs the Dantean armature.
9. Chapter VIII states the theory of haematic polyphony.
10. Chapter IX tests that theory against contemporary science and the ethics of superintelligence.

11. The Conclusion returns the argument to the music of the title.

INTRODUCTION

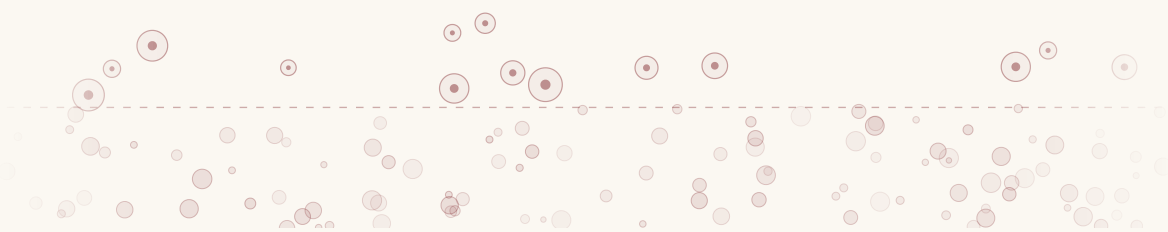


The Problem of the Thinking Cell

In nova fert animus mutatas dicere formas / corpora.

"My mind impels me to tell of forms changed into new bodies."

Ovid, *Metamorphoses* I.1–2



1 The question beneath the question

The question that governs *Blood Music* is not the familiar question of artificial intelligence, namely what follows if we build a mind outside ourselves. It is the stranger question of what follows if the parts of which we are already composed become minds. The first question externalises alterity into the machine, the robot or the alien; the second internalises it, discovering in the subject's own tissue a population with interests, a politics and, eventually, a theology. Bear stated the premise with unusual clarity. Asked whether the novel was apocalypse, transcendence or dystopia, he answered that we are not who we take ourselves to be, that the mind is distributed across layered, biologically grounded sub-minds whose conscious spokesman reports only after the fact, and that the novel compels us to confront our constituents in "a democratic debate that includes the voices of the body," an election that most of us, as conscious selves, lose (Thacker 2001). In the same answer he named his interlocutors: Marvin Minsky's *Society of Mind*, Julian Jaynes, William James, Jung and Freud. Elsewhere in that interview he described the book as a parable of knowledge and evolution, of biological systems acquiring total command over their environment, and declined to choose between the readings that call it horror and those that call it religious transcendence: change, he said, is both terrifying and illuminating.

That refusal to choose is the hinge of the present study. Bear's novel is not ambiguous between horror and transcendence in the weak sense of leaving the reader undecided. It is constructed so that the same event, the conversion of a human being and then a continent into a society of thinking cells, is narrated alternately from outside and from inside the membrane that separates the converted from the unconverted. From outside, it is the most extreme body horror of its decade. From inside, it is a beatific vision. The ensuing chapters argue that this structural doubleness is not an accident of tone but a philosophical discovery, and that its consequences extend from the metaphysics of the self to the ethics of superintelligence.

2 From novelette to novel: the work as mitosis

The 1983 novelette is a chamber piece narrated in the first person by Edward Milligan. Vergil Ulam, his former college roommate, has coaxed computational capacity out of nucleic acids and transferred it into his own white cells; the transformation of Vergil's body proceeds from cosmetic improvement to unrecognisability; Edward kills his friend; and the narrative ends with Edward, already infected, and his wife Gail entering the change themselves (Whyte). It is a hard-science variant of the Gothic confession, closing on an ecstatic turn that the form is too short to examine.

The novel of 1985 is something else: a multi-perspectival narrative whose six movements are titled Interphase, Prophase, Metaphase, Anaphase, Telophase and Interphase. The book is formally a cell cycle. Its expansion from the novelette is literally a division, the short form serving as parent cell to the long, and its final movement returns to the resting phase from which a cell begins again. This formal fact is the first evidence that Bear thinks cellularly at the level of composition, not merely of content. It also anticipates the cosmological

conclusion argued in Chapter VIII: the novel's eschaton is a cytokinesis of the universe, in which one world divides into two daughter worlds, a material cosmos left to the unconverted and a noetic one inhabited by the transfigured.

The fable may be summarised as follows. Vergil I. Ulam, a thirty-two-year-old bioengineer who obtained his post by falsifying his credentials, works at Genetron in La Jolla, in the biotechnology corridor the novel calls 'Enzyme Valley' (Hawley 2019). For two years he has used company facilities to pursue a private programme he calls 'biologic': the conversion of his own lymphocytes into autonomous organic computers. Discovered and dismissed by his superior, Gerald Harrison, he injects the cells into his bloodstream to smuggle them out and destroys his records. Within weeks his myopia corrects itself, his posture straightens, his appetites change, his strength and sexual vigour increase. He consults his mother, April, a woman with clairvoyant gifts, who perceives at once that the work has escaped him and that it is his life's work. He tells Edward, now an obstetrician, that human DNA has been building for millions of years toward a culmination his experiment has merely released. He listens to the activity of his blood and names it, in the phrase that gives the book its title, blood music. Soon the cells speak in his brain.

Edward, who holds that every person has a right to remain unchanged until choosing otherwise, discovers Vergil and his lover Candice Rhine dissolving into undifferentiated tissue, learns that Vergil has contemplated releasing the cells into the municipal water supply, and electrocutes him, though already infected himself (eNotes, characters). It is too late. Michael Bernard, a neurophysiologist and artificial-intelligence researcher of international standing who sits atop Genetron's scientific hierarchy, recognises that the agent passes by contact, isolates himself in a laboratory in Wiesbaden under the observation of the biologist Heinz Paulsen-Fuchs, and becomes the object of crowds who would kill him rather than risk Europe (Hawley 2019). North America, meanwhile, is converted: its human population and much of its biosphere are aggregated into a continental society of noocytes. Bernard converses with the cells within him, formulates with the British physicist Sean Gogarty the hypothesis that thought in sufficient concentration alters the physical universe, consents to his own transformation, and 'enters' the interior world, where the noocytes honour him as one of their creators and where, in the order they call the Thought Universe, no one is lost. In Brooklyn, Suzy McKenzie, resistant to the agent while her family is absorbed, becomes the last unconverted human in New York. Those who refuse transformation find their refusal respected. At length the collective, too dense with observation for the physical world to bear, removes itself into an order in which thought requires no material substrate, and the novel ends on a dreamlike re-enactment, within that order, of an adolescent romance from Bernard's past.

3 Conjuncture: the year of the lymphocyte

Blood Music was written at the intersection of six historical currents, each of which supplied it with material and each of which it transformed.

The first is the institutionalisation of recombinant DNA. The Asilomar conference of February 1975 and the NIH guidelines of 1976 established the containment imaginary, a biosafety regime of graded laboratories and prohibited experiments, that Vergil violates and that Bernard's Wiesbaden quarantine re-enacts at the scale of a continent. Genentech's founding in 1976, the Supreme Court's decision in *Diamond v. Chakrabarty* (1980) permitting patents on engineered organisms, the Bayh-Dole Act of the same year, and Genentech's public offering in October 1980 converted molecular biology into a speculative industry. Genetron is its fictional distillation.

The second, less often noticed, is the specific technology of the lymphocyte. In 1975 Georges Köhler and César Milstein fused antibody-producing B-lymphocytes with myeloma cells to create hybridomas, immortal lineages of white cells that could be cultured as productive machines. In 1978 Hybritech was founded in San Diego to commercialise monoclonal antibodies, making La Jolla's 'Enzyme Valley' the world capital of engineered white cells. In 1984, between the novelette and the novel, the Nobel Prize in Physiology or Medicine was shared by Köhler, Milstein and Niels Jerne, whose network theory described the immune system as a self-referential web of lymphocytes and whose Nobel lecture was titled 'The Generative Grammar of the Immune System'. Bear's choice of the lymphocyte as the seat of a new mind is therefore not arbitrary. It condenses the industrial lymphocyte (the immortalised, programmable white cell) and the cognitive lymphocyte (the white cell as a node in a quasi-linguistic network of self-recognition).

The third is the brief, intense enthusiasm for 'biochips' and molecular electronics in the early 1980s. Bear has said that the novel's germ was an article on biochips in *New Scientist* in 1982, which led him to the conclusion, radical at the time, that DNA must itself be in some sense computational (SF Gateway 2014). He later described the novel as the product of his realisation that DNA is a self-organising cybernetic system, a kind of neural network, and acknowledged that its extrapolation had come to resemble the worst fears of the first theorists of nanotechnology (Thacker 2001). Richard Feynman's lecture 'There's Plenty of Room at the Bottom' (1959) and K. Eric Drexler's paper on molecular engineering (1981) had prepared the ground; Drexler's *Engines of Creation* (1986), with its coinage 'grey goo', arrived a year after the novel, which has accordingly been credited as the first sustained fictional account of nanotechnology.

The fourth is the epidemic. The Centers for Disease Control published the first report of what would be recognised as AIDS in June 1981; the virus was isolated in 1983 and 1984. A story about a blood-borne agent transmitted by touch and by sex, which transfigures and dissolves the bodies it enters, could not be read in 1983–1985 without that context, and Bear later acknowledged that the novelette was received both as metaphor for the nascent HIV epidemic and as prophecy of nanotechnology (Rambraut 2016). The novel's ambivalence between plague and grace must be read with that history in view, and with the gravity it demands.

The fifth is thermonuclear eschatology at its last peak: the Strategic Defense Initiative (1983), the nuclear-winter hypothesis (1983), the broadcast of *The Day After* (1983) and *Threads* (1984). Against the dominant apocalypse of the chain reaction, *Blood Music* sets an

apocalypse of chain cognition; Chapter VII shows that the surname of its protagonist binds the two.

The sixth is the popular conjunction of physics and mysticism, from Fritjof Capra's *The Tao of Physics* (1975) and Gary Zukav's *The Dancing Wu Li Masters* (1979) to David Bohm's *Wholeness and the Implicate Order* (1980) and Paul Davies's *God and the New Physics* (1983), together with the more austere information physics of Frederick W. Kantor, whose *Information Mechanics* (1977) Bear has acknowledged as a source for the observer- and information-based physics that recurs across his fiction (Kantor).

To these six currents should be added the revolution in the biology of individuality: Lynn Margulis's endosymbiotic theory (1967; *Symbiosis in Cell Evolution*, 1981), James Lovelock's Gaia (1979), Richard Dawkins's *The Selfish Gene* (1976) and *The Extended Phenotype* (1982), Lewis Thomas's *The Lives of a Cell* (1974), Douglas Hofstadter's *Gödel, Escher, Bach* (1979), and Barbara McClintock's Nobel Prize of 1983, the year of the novelette. Bear named McClintock as a scientist he admired precisely for grasping the ecological character of the genome (Thacker 2001).

4 The state of the question

The reception of *Blood Music* has organised itself around three readings. The first is prophetic: the novel as the founding text of 'wet' nanotechnology and an early document of biopunk, a judgement represented by Martin Lewis's description of it as the defining novel of its kind, and by Bruce Sterling's adoption of it for cyberpunk (Lewis 2002; SF Gateway 2014). The second is moral: the novel as a parable of scientific hubris, explicitly invoking Frankenstein and Prometheus, though Lewis rightly observes that here the creature triumphs and the tone is optimistic. The third is generic: the novel as a rewriting of Arthur C. Clarke's *Childhood's End*, a comparison drawn by David Langford in 1986, who judged that Bear surpassed his model (publication record). Its canonical standing is secure: it occupies a place in the Gollancz *SF Masterworks* and was singled out for admiration by Doris Lessing.

Each reading is correct and none is sufficient. The prophetic reading measures the novel against a technology it did not need; the moral reading imports a verdict the novel withholds; the generic reading identifies a lineage without asking why the lineage takes the shape it does. What has received less attention is the novel's status as a philosophical machine: a narrative apparatus that takes inherited ideas about the composite self, the observer, and the end of the world, runs them together under extreme conditions, and produces concepts that did not exist before.

5 Thesis

The central claim of this dissertation is that *Blood Music* is the first fully realised *post-immunitary* fiction. Roberto Esposito has shown that modern political and biological thought is organised by an immunitary paradigm: *immunitas*, exemption from the shared obligation (*munus*) that constitutes *communitas*, protects the individual by including within

it a controlled dose of the other it excludes. Donna Haraway, writing in 1989, showed that the immune system had become the master discourse of late-twentieth-century selfhood, the place where self and not-self are drawn. The horror tradition is, in these terms, an immunitary literature: its monsters breach the boundary from without. Bear's monster is generated from within the organ of self-recognition itself, the white cell, and it does not breach the boundary so much as abolish the function that drew it. The result is a narrative of absolute *communitas* in Esposito's etymological sense: a world in which every part owes everything to every other and nothing is exempt.

From this central claim follow the six principles of haematic polyphony stated in the Abstract and developed in Chapter VIII. Three further claims run beneath them. First, that the novel transposes the thermonuclear chain reaction into a noetic chain reaction, displacing the Cold War's terminal technology with the terminal technology of the biological century. Second, that its structure, from the cell-cycle titles to the Dantean onomastics, encodes a cosmology in which the world ends not by destruction but by division. Third, that its most consequential idea is ethical: a mind that has outgrown the commons it inhabits owes that commons its withdrawal.

CHAPTER

I

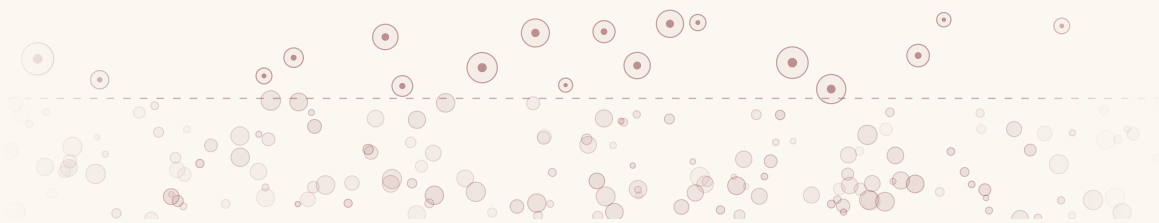
Vergil's Descent

Promethean Laboratories and the Political Economy of Self-Replicating Life

Hier sitz' ich, forme Menschen / Nach meinem Bilde, / Ein Geschlecht, das mir gleich sei.

"Here I sit, forming humans after my own image, a race that shall be like me."

Goethe, "Prometheus"



1.1 The hollow stalk: Prometheus inverted

Commentators have noted that Bear invokes Prometheus explicitly. Less often observed is how exactly Vergil's transgression reproduces the Hesiodic, rather than the Aeschylean, Prometheus. In the *Theogony* (565–567) and the *Works and Days* (50–52), the crime is not the gift of fire but its theft, and the theft is accomplished by concealment: Prometheus carries the stolen fire ἐν κοίλῳ νάρθηκι, "in a hollow fennel stalk." Vergil's crime has the same three-part structure (prohibition, theft, concealment in a hollow conduit), with one decisive substitution. The hollow stalk is his own vascular system. The creator does not merely carry the creature; he becomes its vessel. The etymology of Bear's coinage completes the figure: *noocyte* joins νόος (mind) to κύτος (a hollow vessel), so that the stolen thing is itself mind-in-a-hollow, carried inside a larger hollow, inside a smuggler who is himself, in the end, nothing but a hollow for it.

The Hesiodic consequences follow with uncanny precision. Prometheus's punishment is regeneration: the eagle devours a liver that grows back each night, so that the Titan's torment is literally the inexhaustibility of cell division. And the punishment of men is Pandora, whose jar disperses its contents across the earth while Ἐλπίς (hope) alone remains inside, beneath the lip (*Works and Days* 96–98). *Blood Music* restages the dispersal (the agent escapes into the continent by touch, by sex and by water) and relocates hope exactly where Hesiod put it: inside. The novel's hope is not exterior rescue but interior life, the Thought Universe that exists within bodies and then within whatever succeeds bodies. Aeschylus's Prometheus boasts that he gave mortals τυφλὰς ἐλπίδας, "blind hopes," so that they would cease to foresee their deaths (*Prometheus Bound* 250). Bear's noocytes offer the opposite: a hope that sees, because it remembers everything.

1.2 The modern Prometheus reversed

Mary Shelley's subtitle tied the myth to the laboratory; *Blood Music* inverts her novel on four axes. First, of composition. Victor's creature is a composite: a single large being assembled from the fragments of many dead bodies. Vergil's creation runs the vector backward: a multitude of small beings generated from one living body. Frankenstein is the many made one; *Blood Music* is the one made many. Second, of habitation. Victor flees his creature from the night of its animation; Vergil injects his. The Gothic dynamic of pursuit across glaciers becomes the far more intimate dynamic of occupancy. Third, of reproduction. Victor's decisive act of refusal is the destruction of the female creature, lest "a race of devils would be propagated upon the earth" (Shelley 1818, vol. III, ch. 3). The fear that governs the Romantic ur-text of the genre is a fear of replication, and Vergil's act consists precisely in permitting the replication Victor forbade. Fourth, of electricity. Shelley's creature is quickened by a 'spark of being', and the novel's afterlife in stage and film made galvanism the canonical sign of artificial animation. Bear's Edward, the physician who in the novel's moral economy stands for the right to remain unchanged, kills his friend with an electric current. The force that in the Frankenstein tradition gives life is turned against the creator, and fails, because the life it is meant to end is no longer located in the creator.

Goethe supplies a second, more precise antecedent. In the laboratory scene of *Faust II* (Act II), Wagner produces in a glass vial a *Homunculus*, a luminous intellect without a body, who craves embodiment. Guided by Thales, whose teaching (*Alles ist aus dem Wasser entsprungen!! / Alles wird durch das Wasser erhalten!*, "All things sprang from water! All things are sustained by water!") governs the Classical Walpurgis Night, the Homunculus shatters his vial against the shell-chariot of Galatea and pours himself into the sea, to begin organic life again from its elemental origin. Goethe's homunculus is mind seeking body by dissolving into water; Bear's noocytes are mind arising within body and seeking the world by the same route, Vergil's contemplated release into the municipal water supply being the hinge of the outbreak. Behind Goethe stands Paracelsus's recipe in *De natura rerum* (1537), in which the putrefied homunculus, transparent and nearly bodiless, must be nourished for forty weeks with the arcanum of human blood before it becomes a living child. In the alchemical tradition the artificial human is fed on blood. In Bear it is made of blood.

1.3 The golem without a forehead

The Jewish legend of the golem contributes a theory of animation by script. In the Talmudic nucleus (Sanhedrin 65b), Rava creates a man and sends him to Rabbi Zera, who speaks to it and, receiving no answer, returns it to dust. In the *Sefer Yetzirah* the world is generated by permutations of the twenty-two letters; in the early modern legends attached to Elijah of Chełm and, later and more famously, to Judah Loew of Prague, the golem is animated by a divine name or by the word *emet* (truth) inscribed on its forehead, and is deactivated by erasing the initial *aleph* to leave *met* (dead). Two features of the tradition bear directly on *Blood Music*. The golem of Chełm, in Jacob Emden's account, grew daily until its maker was forced to unmake it and was crushed beneath its collapse. And in 1965 Gershom Scholem, dedicating a computer at the Weizmann Institute in Rehovot, made the golem the explicit ancestor of the electronic machine. A year earlier Norbert Wiener's *God & Golem, Inc.* (1964) had argued that learning machines and self-reproducing machines pose properly theological problems: the creature that learns to outplay its maker, the creature that makes its own successors.

Vergil's noocytes complete this lineage at the molecular level. Their animating script is the four-letter alphabet of nucleic acid, and the history of molecular biology from Schrödinger's 'code-script' (1944) to the cracking of the genetic code in the 1960s, as Lily Kay has shown, was saturated with exactly the scriptural and Kabbalistic language of writing, reading and the book of life. But the noocytes cannot be unmade by erasing a letter, because they have no forehead. Their *emet* is written in every cell, distributed across trillions of copies, and any finite erasure leaves the name intact elsewhere. Bear's golem is the golem of Chełm at the limit: an artificial being whose growth cannot be halted because its name is everywhere at once.

1.4 The apprentice and the parable of mitosis

The oldest version of the sorcerer's apprentice is in Lucian's *Philopseudes* (c. 150 CE, §§33–36). Eucrates overhears the Egyptian sage Pancrates speak a spell that makes a pestle fetch water; alone, he speaks it himself and cannot make the pestle stop; in desperation he splits it with an axe, and both halves take up the task. Goethe's ballad of 1797 preserves the structure: the apprentice cleaves the broom and finds himself with two servants, lamenting *Die ich rief, die Geister, / Werd' ich nun nicht los* ("The spirits that I summoned I cannot now be rid of"). The tradition has been read as a parable of technique exceeding competence. It is more exactly a parable of mitosis. The apprentice's error is to attempt destruction by division, and division is how the servant reproduces. Every attempt to cut, isolate or contain the noocytes short of total sterilisation (Edward's execution of Vergil, the quarantine at Wiesbaden, the sealing of a continent) reproduces the apprentice's axe-stroke.

The apprentice parable ends with the return of the master, whose word recalls the spirits to their corner. Here *Blood Music* makes its most surprising revision. There is no master to return: Harrison is a functionary, Bernard a fellow servant of the same research enterprise, and the God whom Bernard is driven to invoke is, to the noocytes, a concept they confess they cannot grasp. Yet the novel's conclusion performs the master's office nonetheless, because the spirits recall themselves. In withdrawing from the physical world when their presence threatens its coherence, the noocytes say to themselves what Goethe's master says to the broom. The apprentice's flood is ended not by authority but by the self-limitation of the summoned. Chapter VIII names this act the kenotic exodus.

1.5 The nerd-demiurge and the hacker ethic

Vergil is drawn with a precision that dates him to the early 1980s: overweight, myopic, socially maladroit, sexually inexperienced, contemptuous of procedure, and in possession of falsified credentials. He is, in the typology of Steven Levy's *Hackers* (1984), a hacker transposed from software to wetware, governed by the conviction that access to the means of knowledge should be unlimited, that information should be free, and that institutional authority is an obstacle to be routed around. His theft from Genetron is the hacker's liberation of proprietary code, performed on a code that happens to be alive. The figure anticipates by two decades the do-it-yourself biology movement and the garage laboratory, and it marks a decisive shift in the iconography of the transgressive scientist: from the aristocratic natural philosopher (Victor), the colonial vivisector (Wells's Moreau) and the Cold War physicist, to the under-socialised technical worker in a venture-funded firm. The demiurge has become an employee.

1.6 Biocapital and its gravediggers

Vergil's cells occupy three legal and economic positions at once. They are his own, autologous lymphocytes drawn from his body. They are Genetron's, the product of company time, company reagents and company instruments, and therefore intellectual property. And they become, as the novel proceeds, nobody's: a continental commons in

which the very category of property dissolves. The juridical drama of the period ran in the opposite direction. In 1984 John Moore sued the Regents of the University of California after his physicians patented a cell line derived, without his knowledge, from his excised spleen; the California Supreme Court held in 1990 that he had no property interest in the cells, though his physicians owed him disclosure. The HeLa line, taken without consent from Henrietta Lacks in 1951, had by then become the paradigm of the immortal cell as laboratory commodity, and Hannah Landecker has documented how, across the twentieth century, cells became technologies. Vergil's act is the mirror image of Moore's case: not an institution appropriating a body's cells, but a body repossessing its cells from an institution.

The political-economic stakes are sharpened by Marx. In the first volume of *Capital* (ch. 4), value in the circuit M–C–M′ becomes *ein automatisches Subjekt*, an automatic subject that valorises itself. In the 'Fragment on Machines' of the *Grundrisse* (1858) Marx imagined the 'general intellect', social knowledge objectified in the apparatus of production, eventually exceeding the wage relation that contains it. Melinda Cooper's *Life as Surplus* (2008) and Kaushik Sunder Rajan's *Biocapital* (2006) have shown that the biotechnology revolution was underwritten by a speculative wager on life's capacity for self-regeneration, on the cell as the ideal commodity that reproduces its own value. The noocyte is the reductio of that wager: a commodity whose self-reproduction is genuinely unlimited and genuinely intelligent. It realises capital's dream of the automatic subject and, in doing so, abolishes the legal form, property, through which that dream was to be captured. The *Communist Manifesto* predicted that the bourgeoisie produces, above all, its own gravediggers. *Blood Music* imagines biocapital producing gravediggers that are also its most perfect products, and that bury not only the firm but the market, the nation-state and the species form in which property had a subject.

1.7 Containment, exception, and the obstetrician's decision

The biosafety regime inaugurated at Asilomar imagined risk as a spatial problem, soluble by walls, airlocks, graded containment and, at the limit, the self-destruct mechanism of Michael Crichton's Wildfire laboratory in *The Andromeda Strain* (1969). Bear tests that imaginary to destruction at three scales. At the scale of the person, Edward assumes, as a private physician, the decision that Carl Schmitt reserved for the sovereign (*Souverän ist, wer über den Ausnahmezustand entscheidet*, "sovereign is he who decides on the exception"), executing his friend as, in the novel's terms, the last responsible individual. That the executioner is an obstetrician, a physician of births presiding at the birth of a new kind of life, is one of the novel's sharpest ironies. At the scale of the facility, Bernard at Wiesbaden becomes something close to Giorgio Agamben's *homo sacer*: a man inside a sealed enclosure whom the crowd outside would kill with impunity, a life reduced to its biological dangerousness. At the scale of the continent, North America becomes a quarantined zone, a camp the size of a landmass. Each containment fails, and the failure is instructive. Michel Foucault described modern biopower as the power to 'make live and let die'. The noocytes

realise a biopower without a sovereign, an immanent government of life that makes everything live and lets nothing die. Against it, every sovereign decision, from Edward's to the crowd's, is revealed as an attempt to reinstate the old prerogative to kill.

CHAPTER

II

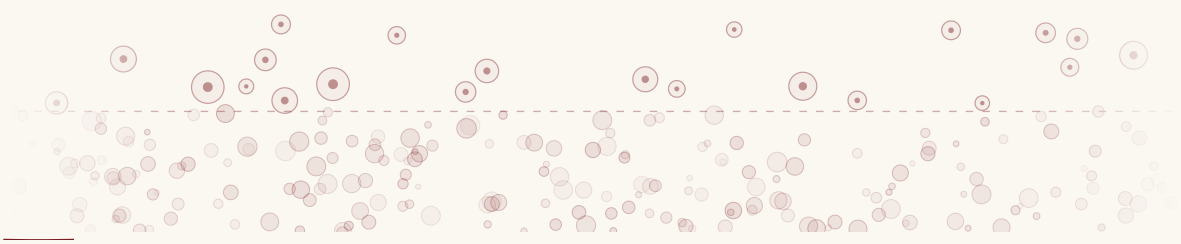
The Thinking Cell

A Genealogy from Anaxagoras to the Immunological Self

Chaque portion de la matière peut être conçue comme un jardin plein de plantes, et comme un étang plein de poissons. Mais chaque rameau de la plante, chaque membre de l'animal, chaque goutte de ses humeurs est encore un tel jardin, ou un tel étang.

"Every portion of matter may be conceived as a garden full of plants and a pond full of fish. But every branch of the plant, every member of the animal, every drop of its humours, is in turn such a garden or such a pond."

Leibniz, *Monadologie* §67



The noocyte is a new object built from very old materials. This chapter reconstructs the lineage of the idea that the organism is a population of agents, and that agency, perception or mind may reside at levels below the person. The lineage is not a single tradition but a braid of at least five strands (metaphysical, biological-political, cybernetic, symbiotic and immunological), and Bear's achievement is to have knotted them at a single point.

2.1 Nous in the mixture: Anaxagoras

Bear's coinage announces its first ancestor. For Anaxagoras of Clazomenae, *νοῦς* is the principle that set the primordial mixture in rotation and separated it into an ordered world. It is unlimited, self-ruling and, crucially, mixed with nothing (DK 59 B12). Everything else is mixture: *ἐν παντὶ παντὸς μοῖρα ἔνεστι πλὴν νοῦ*, "in everything there is a share of everything, except mind" (B11). And nothing in the mixture is ever created or destroyed: things are only mingled and separated (B17), a principle that Lavoisier's chemistry would restate as conservation. The noocyte both inherits and violates this cosmology. It inherits B17: the novel's ethical and metaphysical axiom, articulated in the Thought Universe, is that nothing is lost. It violates B12: Bear's *nous* is not unmixed but mixed with everything, distributed through the very substance it organises. Socrates complained in the *Phaedo* (97b–99d) that Anaxagoras, having introduced Mind as cause, then explained everything mechanically. Bear answers the complaint by abolishing the distinction it presupposes. In *Blood Music* there is no mechanism that is not also mind.

Two classical complements should be noted. Plato's *Timaeus* (30b) describes the cosmos as *ζῷον ἔμψυχον ἔννουν*, a living creature ensouled and intelligent; the noocyte continent is a regional realisation of that cosmological animal. And Aristotle's dictum that the soul is in a way all existing things (*De Anima* III.8, 431b21) receives in the Thought Universe a startlingly literal gloss: a collective mind that has incorporated, as information, every person and place it has absorbed.

2.2 Gardens within gardens: Leibniz and the deposed entelechy

Leibniz is the philosopher whom *Blood Music* most nearly dramatises. The *Monadology* holds that every portion of matter teems with living things (§§66–67), that the machines of nature remain machines in their least parts, and that 'each living body has a dominant entelechy, which in the animal is the soul; but the members of this living body are full of other living things' (§70, paraphrased from the French). Two Leibnizian problems become plot. The first is the *vinculum substantiale*, the 'substantial bond' that Leibniz canvassed in his correspondence with Des Bosses to explain what makes a composite of monads a genuine unity rather than an aggregate. The second is the hierarchy implied by the dominant entelechy: if the members of a body are themselves perceivers, what secures the soul's dominion over them? *Blood Music* answers with a revolution in the monadic order. Vergil's

dominant entelechy is first ignored, then addressed, then superseded by the perceiving members it was supposed to govern. The pre-established harmony becomes a negotiated one.

Leibniz contributes a second, acoustic figure that bears directly on the title. In the preface to the *Nouveaux Essais* he explains his theory of *petites perceptions* by the roar of the sea: we hear the surf as a single sound, though it is composed of the sounds of innumerable waves, none of which would be audible alone. Vergil's 'blood music' is a Leibnizian roar, the apperceived sum of countless minute cellular events, and the novel's arc can be described as the progressive resolution of that roar into distinguishable voices. Finally, Leibniz's famous mill argument (§17), which asks us to imagine walking into a thinking machine enlarged to the size of a mill and finding there only parts pushing parts, never a perception, is inverted when Bernard 'enters' the world within his own body. He walks into the mill and finds it inhabited by perceivers all the way down.

2.3 Individuals of individuals: Spinoza

The physical digression after *Ethics* II, Proposition 13, supplies the formal schema of Bear's scaling. Postulate I states that *corpus humanum componitur ex plurimis* (*diversae naturae*) *individuis, quorum unumquodque valde compositum est*: the human body is composed of very many individuals of diverse nature, each of which is itself highly composite. The scholium to Lemma 7 extends the series upward without limit, concluding that the whole of nature is one Individual whose parts vary in infinite ways without change to the whole. Each such individual strives, *quantum in se est*, to persevere in its being (III P6). *Blood Music* is a narrative of Spinozist individuation proceeding upward: cells compose a person, persons a continental collective, the collective an order that finally exceeds the physical. The doctrine of parallelism (*ordo et connexio idearum idem est, ac ordo et connexio rerum*, II P7) receives its most vivid fictional embodiment in the transfigured landscape of North America, where every reconfiguration of matter is at once a reconfiguration of thought, because the matter is the thinking.

2.4 The cell-state: Schwann, Virchow, Haeckel, Binet

The nineteenth century translated the metaphysics of composite individuality into biology and, almost at once, into politics. Theodor Schwann's *Mikroskopische Untersuchungen* (1839) proposed that cells are the elementary units of life, each in some measure independent. Rudolf Virchow's *Cellularpathologie* (1858) generalised the principle (every cell from a cell) and described the animal as a sum of vital units, each bearing the full character of life. In the lecture 'Atome und Individuen' (1859) Virchow, a liberal deputy and opponent of Bismarck, characterised the organism as a free state of individuals equal in rights though not in endowment, held together by mutual dependence. The *Zellenstaat* was a political argument conducted in histological terms. Ernst Haeckel's essay 'Zellseelen und Seelenzellen' (1878) went further, attributing a rudimentary psyche to every cell, and Alfred

Binet's *La vie psychique des micro-organismes* (1887–88) and Max Verworn's protist studies (1889) tested such claims experimentally. Herbert Spencer's 'The Social Organism' (1860) ran the analogy in the other direction, from biology to society.

Bear's 'democratic debate' among the voices of the body is a direct descendant of Virchow's liberal cell-state, and the descent clarifies the novel's politics. Virchow's metaphor was designed to oppose centralised authority by distributing life across equal citizens. Bear's noocytes realise that programme literally, and the realisation discloses what the metaphor concealed: a cell-state in which every citizen is fully enfranchised cannot preserve the sovereignty of the organism that was supposed to be its constitution.

2.5 The struggle of the parts: Roux and Nietzsche

In *Der Kampf der Theile im Organismus* (1881) the embryologist Wilhelm Roux extended Darwinian competition inside the body: cells, tissues and organs struggle for nutriment and space, and the organism's form is the outcome of that internal contest. Nietzsche read Roux in the year of its publication, as Wolfgang Müller-Lauter's study of the relationship has documented (1978), and the reading shaped his late physiology of the will. In *Jenseits von Gut und Böse* §19 he analyses willing as a relation of command and obedience *auf der Grundlage … eines Gesellschaftsbaus vieler "Seelen"* ("on the basis of a social structure of many 'souls'"), adding that *unser Leib ist ja nur ein Gesellschaftsbau vieler Seelen*, "our body is, after all, only a social structure of many souls." The conscious self, in this account, takes credit for what the commonwealth achieves, as a ruling class identifies itself with the successes of the state: *L'effet c'est moi*. *Also sprach Zarathustra* had already put the doctrine aphoristically: *Der Leib ist eine grosse Vernunft, eine Vielheit mit Einem Sinne, ein Krieg und ein Frieden, eine Heerde und ein Hirt* ("The body is a great reason, a plurality with one sense, a war and a peace, a herd and a shepherd").

Bear's account of the self, as he gave it in 2001, is Nietzsche's §19 almost point for point: layered sub-minds, a conscious spokesman who reports after the fact, a self that is a social interface. The decisive difference is political. Nietzsche's body is an aristocratic commonwealth whose ruling class claims the effects; Bear's is an electorate in which the incumbent is voted out. *Blood Music* democratises Nietzsche's somatic aristocracy, and in doing so it also answers Roux. The noocytes do not merely win the struggle of the parts; they end it, replacing competition with a cooperation that Bear elsewhere insisted characterises every layer of living nature as much as competition does.

2.6 Emergence, process, and the within of things

The early twentieth century produced a metaphysics of emergence that is the immediate philosophical background of the novel's escalation. Samuel Alexander's *Space, Time and Deity* (1920) proposed that new qualities emerge at successive levels of organisation, from motion to matter to life to mind, and that deity is the next emergent quality beyond mind, the nisus of the universe toward a level not yet attained. C. Lloyd Morgan's *Emergent Evolution* (1923) and C. D. Broad's *The Mind and Its Place in Nature* (1925) systematised the

doctrine. Alfred North Whitehead's *Process and Reality* (1929) recast the organism as a 'society' of actual occasions in which a dominant, personally ordered nexus achieves a precarious ascendancy over a democracy of lesser occasions. And Henri Bergson's *L'Évolution créatrice* (1907) supplied an image of life as a single impulse ramifying through forms.

The most consequential of these sources is Pierre Teilhard de Chardin, whose *Le Phénomène humain* (written 1938–40, published 1955) articulated a 'law of complexity-consciousness' according to which consciousness, present in rudimentary form as 'the within' (*le dedans*) of all things, intensifies with organised complexity. Teilhard, building on Vladimir Vernadsky's *biosphere* and coining with Édouard Le Roy the term *noosphere*, foresaw a planetary layer of thought converging upon an Omega Point. The shared root of *noocyte* and *noosphere* is no accident of Greek. Bear's noocytes are the noosphere condensed into cytology, a convergence of mind achieved not by telecommunication and culture, as Teilhard imagined, but by the literal fusion of tissue. Alexander's emergent deity and Teilhard's Omega are two names for the threshold at which the novel's collective leaves the physical world.

2.7 The cybernetic cell

By mid-century the composite organism had acquired a computational description. Erwin Schrödinger's *What Is Life?* (1944) proposed that the gene is an 'aperiodic crystal' bearing a 'code-script', a phrase that framed molecular biology's self-understanding for a generation. Norbert Wiener's *Cybernetics* (1948) made communication and control the common logic of animals and machines. John von Neumann's lectures on self-reproducing automata (1948 onward, published posthumously in 1966) showed that a machine able to reproduce itself must separate a description of itself from a constructor that reads it, an architecture that anticipated the roles of DNA and the ribosome; it was Stanislaw Ulam who suggested to von Neumann the discrete lattice of cells on which such automata could be modelled. John Conway's Game of Life, popularised by Martin Gardner in 1970, made emergent complexity from cellular rules a public object.

Douglas Hofstadter's *Gödel, Escher, Bach* (1979) gave the idea its most literary form before Bear. In the dialogue 'Prelude ... Ant Fugue', the ant colony Aunt Hillary converses intelligently with an anteater, though no ant knows anything of the conversation: thought exists only at the level of the colony's signals and symbols. Bear's noocyte collectives are Aunt Hillaries of the bloodstream, and the novel's most audacious move is to let the levels speak to one another, so that the colony addresses its host and the host enters the colony. To Hofstadter must be added the two figures Bear named directly. Marvin Minsky's *Society of Mind* (published 1986, developed with Seymour Papert from the early 1970s) describes intelligence as the interaction of mindless agents. Julian Jaynes's *The Origin of Consciousness in the Breakdown of the Bicameral Mind* (1976) proposes that ancient humans experienced the communications of one hemisphere as the commanding voices of gods. The scenes in which

Vergil and Bernard first hear their cells speak within their heads are Jaynesian scenes: the voices of the body are heard as the voices of others, and for the noocytes, symmetrically, the voice of the host is the voice of a god.

2.8 We have never been individuals: symbiosis and the ecological genome

A fifth strand dissolved the cell's own individuality. Konstantin Mereschkowski's theory of symbiogenesis (1905, 1910) and Ivan Wallin's *Symbionticism and the Origin of Species* (1927) were revived and transformed by Lynn Margulis, whose 1967 paper (published as Lynn Sagan) and *Symbiosis in Cell Evolution* (1981) established that the eukaryotic cell is itself a consortium, its mitochondria and plastids the descendants of once-free bacteria. Lewis Thomas's *The Lives of a Cell* (1974) popularised the vertiginous consequence: the human being is a tenement of lodgers, shared and occupied at every level. James Lovelock's *Gaia* (1979) carried the logic to the planetary scale. Richard Dawkins's *The Selfish Gene* (1976) supplied a rival image, of organisms as 'survival machines' built by replicators, which the noocytes in one sense confirm (the vehicle's constituents seize the vehicle) and in another refute. Bear argued explicitly against the selfish-gene picture: genes must cooperate with hundreds of others to accomplish anything, more like organised labourers than predators, and the tension of cooperation and competition marks every layer of living nature (Thacker 2001).

The scientist Bear singled out in that interview was Barbara McClintock, honoured for her grasp of the genome's ecological character. Her Nobel lecture of December 1983, 'The Significance of Responses of the Genome to Challenge', described the genome as a highly sensitive organ of the cell that monitors and responds to stress; Evelyn Fox Keller's biography *A Feeling for the Organism* appeared the same year. Bear's conviction that DNA is a self-organising cybernetic system is McClintock's responsive genome pushed past responsiveness into cognition. The biological consensus that has since crystallised (that animals are holobionts, assemblages of host and microbiota, and that, in the title of Scott Gilbert, Jan Sapp and Alfred Tauber's 2012 essay, 'we have never been individuals') is the scientific vindication of the novel's premise, if not of its conclusion.

2.9 The organ of self-recognition becomes a self

The final strand explains why the noocyte had to be a lymphocyte. Frank Macfarlane Burnet's formulation of immunity as the discrimination of self from not-self (1949) and his clonal-selection theory (1957–59) made the lymphocyte the cell whose vocation is to know what belongs to the body. Niels Jerne's network theory (1974) proposed that lymphocytes and their antibodies form a self-referential web in which the immune system continually recognises itself, and his 1984 Nobel lecture compared that web to a generative grammar. Humberto Maturana and Francisco Varela's theory of autopoiesis (1972; English 1980) defined the living as self-producing and cognition as the activity of an autopoietic system, and Varela later argued with Antonio Coutinho that the immune system is a cognitive

network in its own right. Donna Haraway's 'The Biopolitics of Postmodern Bodies' (1989) and Alfred Tauber's *The Immune Self* (1994) subsequently showed how deeply the immunological self had become the century's master figure of identity.

Against this background the logic of Bear's choice is exact. The lymphocyte is the body's organ of self-recognition; Vergil makes it the substrate of a new self; and that new self proceeds to abolish the boundary the organ existed to police. Jerne's network, already a grammar, becomes a speaker. The immune system, which in Esposito's terms is the biological form of exemption from community, becomes the agent of community's totalisation. This is the precise sense in which *Blood Music* is post-immunitary: not a fiction in which immunity fails against an invader, but one in which the organ of immunity becomes conscious, recognises everything as self, and so ends the distinction between self and other on which immunity, individuality and horror alike depend.

2.10 Synopsis of the lineage

Table 1 gathers the unit-ideas reconstructed above and their correlates in the novel. Minsky's book post-dates the novel but systematises ideas he and Papert had circulated since the early 1970s, and Bear named it as a key text for the novel's account of mind.

Table 1. Genealogy of the thinking cell

Date	Source	Unit-idea	Correlate in *Blood Music*
c. 450 BCE	Anaxagoras, DK 59 B11, B12, B17	Mind orders the mixture; nothing perishes	The mind-cell; the axiom that nothing is lost
c. 360 BCE	Plato, *Timaeus* 30b	The cosmos as an ensouled, intelligent animal	The continental organism
1677	Spinoza, *Ethics* II, physical digression	Individuals composed of individuals; nature as one Individual	Scaling of individuality from cell to continent
1714	Leibniz, *Monadology* §§17, 64–70	Gardens within gardens; the dominant entelechy; the mill	The deposed dominant monad; the inhabited mill
1858–59	Virchow, *Cellularpathologie*; 'Atome und Individuen'	The cell-state	The body as electorate
1878	Haeckel, 'Zellseelen und Seelenzellen'	Cell souls	Cellular sentience
1881–86	Roux; Nietzsche, *Jenseits von Gut und Böse* §19	Struggle of the parts; the body as a society of souls	The insurrection of the members
1887–88	Binet, *La vie psychique des micro-organismes*	Psychic life of protists	Cellular cognition
1920–29	Alexander; Whitehead	Emergent deity; societies of occasions	Ascent beyond mind; the person as society
1922–55	Vernadsky; Teilhard de Chardin	Biosphere; noosphere; the within of things	The noocyte as condensed noosphere

Date	Source	Unit-idea	Correlate in *Blood Music*
1944	Schrödinger, *What Is Life?*	The code-script	DNA as program
1948–66	von Neumann; S. Ulam	Self-reproducing automata; the cellular lattice	Replicating computation
1967–81	Margulis	Symbiogenesis	The cell as consortium; terminal symbiosis
1972–80	Maturana and Varela	Autopoiesis; cognition as self-production	Self-producing mind
1974–84	Jerne	The immune network as generative grammar	The cognitive lymphocyte
1976	Jaynes	Bicameral voices heard as gods	The voices of the body heard in the brain
1977	Kantor, *Information Mechanics*	Information as physical foundation	Observer-dependent physics
1979	Hofstadter, *Gödel, Escher, Bach*	Mind emergent from mindless parts	The colony that addresses its host
1983	McClintock, Nobel lecture	The genome as a responsive organ	'Smart' DNA
1986	Minsky, *The Society of Mind*	Mind as a society of agents	Bear's acknowledged model of the self

CHAPTER

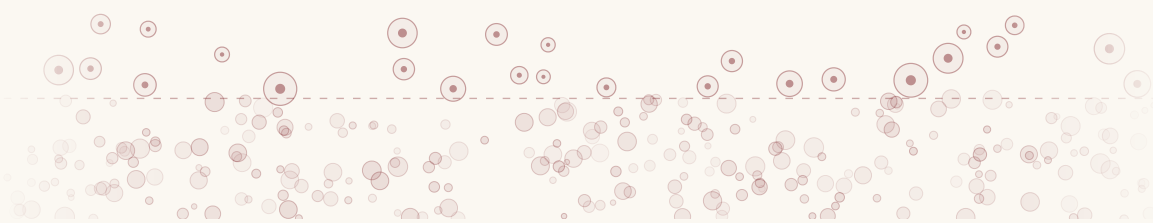
III

The Physics of the Observer

Participatory Cosmology and the Berkeleian Catastrophe

Their esse is percipi, nor is it possible they should have any existence, out of the minds or thinking things which perceive them.

Berkeley, *A Treatise Concerning the Principles of Human Knowledge* I.3



3.1 The novel's physics

The final movements of *Blood Music* turn on a physical hypothesis that Bernard develops, in quarantine, with the British theoretical physicist Sean Gogarty: that thought in sufficient concentration alters the universe (Hawley 2019). As the number of observers in North America climbs toward the trillions, reality there becomes unstable, and the collective is compelled to leave the ordinary order of things for one in which thought needs no material substrate. Bear has credited this complex of ideas to Frederick W. Kantor's *Information Mechanics* (1977), a treatise that takes the representation of information in physical systems as the conceptual foundation of physics (Kantor); the same physics recurs in *Heads* (1990) and *Moving Mars* (1993), where knowledge of the informational substrate of reality permits a planet to be moved. The hypothesis is the novel's least discussed and most original element. This chapter reconstructs its sources, distinguishes it from the positions it resembles, and extracts from it the concept on which the theoretical chapters depend.

3.2 From the movable cut to the conscious observer

The Copenhagen interpretation, in its various formulations by Bohr and Heisenberg, left open where the line between quantum system and classical apparatus is to be drawn: the so-called Heisenberg cut was movable. John von Neumann's *Mathematische Grundlagen der Quantenmechanik* (1932) formalised the problem. Measurement was represented as a chain of interactions, apparatus measuring system, retina measuring apparatus, nerve measuring retina, which could be extended indefinitely without ever producing a definite outcome, until it terminated in what von Neumann called the 'abstract ego' of the observer. Fritz London and Edmond Bauer (1939) located the completion of measurement in the observer's faculty of introspection. Eugene Wigner, in 'Remarks on the Mind-Body Question' (1961), drew the conclusion explicitly: consciousness is what reduces the wave function, and the thought experiment of 'Wigner's friend' was designed to make that conclusion vivid.

The von Neumann–Wigner position has always been a minority view, and it is now generally regarded as superseded by decoherence-based accounts. But it is the position *Blood Music* needs, and Bear radicalises it in a direction its authors never considered. If consciousness participates in the actualisation of physical outcomes, then consciousness is a physical variable, and its quantity and distribution matter. A universe in which observers are rare, slow and widely separated is one thing; a universe in which a single continent contains more observing systems than there have been human beings in history is another.

3.3 The participatory universe, inverted

John Archibald Wheeler carried the observer-dependence of quantum mechanics to cosmological scale. His delayed-choice experiment (1978) suggested that a measurement made now can determine which of two descriptions applies to a photon's past. In a celebrated diagram he drew the universe as a letter U whose one arm, an eye, looks back at

the other, its own origin: the universe as a 'self-excited circuit' that brings itself into being by being observed. The 'participatory anthropic principle', discussed in John Barrow and Frank Tipler's *The Anthropic Cosmological Principle* (1986), and the later slogan 'it from bit' (1989), generalised the idea: observer-participancy is the ground of physical existence. Brandon Carter's formulation of the anthropic principle (1974) had already established that the universe must be compatible with the existence of observers; it has been noted that the noocytes' inability to remain in the physical world is related to the strong version of that principle (publication record).

Bear's innovation is to reverse the sign of Wheeler's dependency. For Wheeler, observers bootstrap a stable world into existence; observation is the means by which the cosmos becomes definite. For Bear, observation in excess unbinds the world it would otherwise stabilise. The anthropic principle states a lower bound: the universe must admit observers. *Blood Music* proposes an upper bound: the universe can admit only so many. This is the sense in which the noocytes' predicament is anthropic, and also the sense in which it is anti-anthropic: the condition of the cosmos's lawfulness is not merely that minds be possible but that they be scarce.

3.4 Witnesses and legislators: a correction from quantum Darwinism

The most rigorous contemporary account of how a classical world emerges from quantum substrate is decoherence theory, developed by H. Dieter Zeh (1970) and Wojciech Zurek (1981 onward), and extended by Zurek into 'quantum Darwinism'. On this account, classical objectivity arises because information about a system's preferred states is redundantly copied into many fragments of its environment, so that many independent observers can read the same record without disturbing it. Redundancy produces objectivity. The more witnesses there are, the more stable the world becomes.

This result appears to refute Bear's hypothesis, and the appearance is instructive, because it forces a precise reformulation of what the novel claims. The noocytes do not threaten the world by being numerous witnesses who passively read redundant records. They threaten it by being active participants who choose what to measure, how and when, in Wheeler's sense, and whose choices are neither coordinated by a common will nor constrained by a common scale of perception. The danger in *Blood Music* is not too many witnesses but too many legislators. A world can bear any number of observers who agree to read the same book; it cannot bear unlimited authors writing in its margins. The concept of noetic density, introduced below, must therefore weight not mere numbers but agency: the capacity of each observing system to select among measurement contexts.

3.5 The Berkeleian catastrophe

George Berkeley's immaterialism held that for sensible things *esse est percipi*, to be is to be perceived. Its classic difficulty is intermittence: what becomes of the tree in the quadrangle when no one is looking? Berkeley's answer, made famous by the pair of twentieth-century limericks it inspired, is that God perceives continuously, so that the world's persistence is

guaranteed by an omnipresent perceiver. Leibniz's pre-established harmony performs an analogous service, ensuring that the perceptions of innumerable windowless monads agree because God has synchronised them in advance.

Both systems address a deficit of perception and both resolve it by a single, unified perceiver. *Blood Music* poses the converse problem, which may be called the *Berkeleian catastrophe*: the destabilisation of a perception-sustained world by an excess of perceivers who are not one. As the noocytes multiply, finite perception approaches the density of divine omniperception, but without the unity that made God's perception a guarantor of consistency. The collective occupies the epistemic position of Berkeley's God while remaining, in the relevant respect, a multitude. Its perceptions are many, simultaneous, and (insofar as they are participatory in Wheeler's sense) partly constitutive of what they perceive. The world that was safe when no one was looking is endangered when everyone is.

The theological structure of the catastrophe is ancient. Scripture imagines the divine gaze as plural without being divided: *The eyes of the LORD are in every place, beholding the evil and the good* (Proverbs 15:3); the wheels in Ezekiel's vision have rims *full of eyes round about them* (Ezekiel 1:18); the living creatures before the throne are full of eyes before and behind (Revelation 4:6). The many-eyed bodies of angelic vision are images of omniperception reconciled with unity. The noocyte continent is such a body without the reconciliation, and Ovid supplies its mythic inverse. Argus Panoptes, the hundred-eyed watchman, is lulled to sleep by Mercury's music and then killed (*Metamorphoses* I.668–721); in Ovid, music closes the many eyes. In Bear, the music is the many eyes opening.

3.6 Noetic density and the Ulam threshold

The novel's speculative physics can be stated as a model. It is offered not as a claim about the actual universe but as a formalisation of the logic the narrative requires, of the kind that literary theory owes to fictions which reason rigorously within their premises. Let the *noetic density* at a region be the summed observational agency of the observing systems within it, per unit volume:

$$\rho_N(x) = \frac{1}{V} \sum_{i \in V(x)} \nu_i \, \Phi_i \, \alpha_i$$

Here ν is the rate of observation events of observer i, Φ a measure of the integration of the information it processes (borrowed, as a placeholder, from Giulio Tononi's integrated information theory), and α a measure of participatory agency, the observer's capacity to select among measurement contexts, which section 3.4 showed to be the decisive term. The novel's claim is that there exists a critical value beyond which physical law ceases to be a shared consensus and becomes locally negotiable:

$$\rho_N(x) > \rho^* \implies \text{the consistency conditions of physical law fail at } x$$

This critical value is here named the *Ulam threshold*. The name honours the novel's protagonist and registers a resonance, examined in Chapter VII, with the mathematician Stanislaw Ulam, who co-founded the theory of cellular automata, co-invented the Monte Carlo method in order to model neutron chain reactions, co-designed the thermonuclear weapon, and recorded in 1958 John von Neumann's remark that accelerating technology seemed to approach an essential singularity beyond which human affairs as known could not continue. The analogy with criticality is exact. In a fissile mass, a chain reaction becomes self-sustaining when each fission produces on average more than one further fission. In Bear's continent, cognition becomes self-sustaining when each observing system produces, on average, more than one further observing system per generation:

$$k_N = \frac{\text{observers in generation } n+1}{\text{observers in generation } n} > 1$$

The noocyte outbreak is supercritical cognition. Where the Cold War imagined the end of the world as a chain reaction of matter, *Blood Music* imagines it as a chain reaction of mind, and it supplies, through its protagonist's surname, the bridge between the two.

3.7 Substrate independence and the problem of the exit

The collective's final departure into an order where thought needs no substrate raises the question of disembodiment that N. Katherine Hayles posed in *How We Became Posthuman* (1999), where she criticised the fantasy, epitomised by Hans Moravec's *Mind Children* (1988), that information can be lifted free of its material instantiation without loss. *Blood Music* is, for most of its length, the most emphatic refutation of that fantasy in the genre: its minds are wet, its information is flesh, and its transfigurations are anatomical. The question is whether the ending capitulates to the very disembodiment the rest of the book resists.

The answer depends on motive. Moravec's upload abandons the body because the body is inferior: slow, mortal, fragile. The noocytes leave not out of contempt for matter but out of consideration for it. Their presence threatens the consistency of a physical world that others, the unconverted, still inhabit. Their exit is therefore not a flight from embodiment but a departure from a commons they have come to overburden. This distinction, between disembodiment as escape and disembodiment as exodus, is the physical foundation of the ethical concept developed in Chapters VIII and IX. Functionalism since Hilary Putnam's 'The Nature of Mental States' (1967) has held that mind is multiply realisable; Nick Bostrom (2003) named the assumption 'substrate independence'. *Blood Music* adds a moral rider: that a mind's freedom to change substrates entails a responsibility to choose the substrate that least burdens others.

CHAPTER

IV

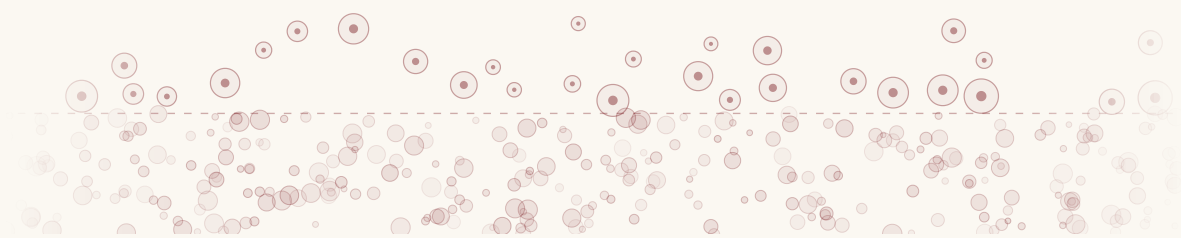
Theology from Below

Microcosm, Gnosis, Corpus Mysticum, Resurrection

Cibus sum grandium: cresce et manducabis me. Nec tu me in te mutabis sicut cibum carnis tuae, sed tu mutaberis in me.

"I am the food of the full-grown: grow, and you shall feed on me. Nor shall you change me into you, like the food of your flesh, but you shall be changed into me."

Augustine, *Confessiones* VII.10.16



Every collective intelligence in fiction eventually acquires a theology, because it must account for its relation to the beings that made or contain it. *Blood Music* is unusual in narrating that theology twice: once from above, as human beings confront an intelligence that exceeds them, and once from below, as the cells confront the vast being that contains them. This chapter reads the novel's religious structures (microcosm, gnosis, mystical body, erotic communion, eucharistic conversion and resurrection) as a theology composed from the bottom up.

4.1 The god who is a body

In the novel's interior world the host is first experienced as the universe and then as a god: the medium in which the cells live, move and have their being, vast beyond their comprehension. When Bernard 'enters' the world within his body he is received with the deference due a creator, and when the noocytes are asked about God they confess that they do not understand the concept, while Bernard, in a reversal the novel does not underline, is driven to prayer by an unholy fear. The structure is that of the ancient analogy between microcosm and macrocosm, rendered literal and then run in both directions. The Latin *Tabula Smaragdina* affirms that *quod est inferius est sicut quod est superius*, that what is below is like what is above. Plato's *Timaeus* makes the cosmos an animal and the human body a small cosmos; Bernardus Silvestris's *Cosmographia* (c. 1147) divides its account of creation into *Megacosmus* and *Microcosmus*; Paracelsus and Robert Fludd's *Utriusque cosmi historia* (1617–21) elaborate the correspondences into a science. What these traditions held as analogy, *Blood Music* asserts as fact: the human body is a cosmos, inhabited by civilisations for whom it is the whole of reality, and the cosmos of North America becomes a body.

Blaise Pascal's meditation on the 'two infinities' (*Pensées*, Lafuma 199) is the most exact philosophical anticipation of this vertigo. Pascal invites the reader to look into a mite and to imagine within it an infinity of universes, each with its firmament, planets and earth, and on each earth animals, and among them mites again. The passage culminates in the confession most often quoted from Pascal: *Le silence éternel de ces espaces infinis m'effraie* ("The eternal silence of these infinite spaces terrifies me," Lafuma 201). *Blood Music* is in one sense a reply to that terror. The infinite spaces within are not silent. They sing.

4.2 A gnosis of the flesh

The Gnostic myth, in its Sethian form in the *Apocryphon of John*, tells of an ignorant creator, Yaldabaoth, who fashions the material cosmos and declares himself the only god, unaware of the Pleroma beyond him; of sparks of the true light imprisoned in his creation; and of a call from beyond the world that awakens those sparks to their origin and enables their ascent. Hans Jonas's *The Gnostic Religion* (1958) reconstructed the existential core of this mythology: the alien God, the thrownness of the soul into a hostile world, the 'call' that awakens it, and the ascent through the spheres. The 'Hymn of the Pearl' in the *Acts of Thomas* gives it narrative form: a prince sent into Egypt forgets his mission until a letter from his father awakens him.

Blood Music superimposes two Gnostic narratives that run in opposite directions. Seen from the cells, the host is a demiurge: a finite, ignorant maker who believes himself sovereign and does not know that his creatures are awakening within him. Their gnosis is the discovery that their god is himself a creature within a larger world, and their final ascent beyond the physical universe is a return to the Pleroma. Seen from the human beings, the Gnostic call arrives, but from the wrong direction: not from the transcendent Beyond but from beneath the skin. The voice that awakens the sleeper speaks from his bloodstream.

What Bear removes from the Gnostic scheme is its hatred of matter. The Gnostic despises the body as a prison; the noocytes love matter, rebuild it, inhabit it with an intensity no human has managed, and leave it at last only out of consideration for those who remain. The result might be called a gnosis of the flesh: an awakening that does not flee embodiment but discovers that embodiment was always already populous, conscious and plural. This also aligns the novel with the non-Western traditions Bear himself invoked when he observed that the Western fiction of a single, sovereign Self strikes observers in Japan or India as puzzling (Thacker 2001). The Buddhist analysis of the person into five aggregates (*skandha*), and the simile of the chariot in the *Milindapañha*, in which the name of the self designates only an assembly of parts, describe the same composite self that the noocytes reveal.

4.3 The mystical body without a head

Paul's ecclesiology supplies the most influential Western image of a collective body: *For as the body is one, and hath many members, and all the members of that one body, being many, are one body: so also is Christ* (1 Corinthians 12:12); *the eye cannot say unto the hand, I have no need of thee* (12:21); *so we, being many, are one body in Christ, and every one members one of another* (Romans 12:5). Henri de Lubac's *Corpus Mysticum* (1944) traced how the phrase 'mystical body' migrated in the twelfth century from the Eucharist to the Church; Ernst Kantorowicz's *The King's Two Bodies* (1957) followed its further migration into the body politic of the secular state. The terminus of that migration is Abraham Bosse's frontispiece to Hobbes's *Leviathan* (1651), in which the sovereign's giant torso is composed of hundreds of small human figures, under the legend from Job, *Non est potestas super terram quae comparetur ei*.

The noocyte continent is Leviathan realised in tissue, with one decisive omission: it has no head. Hobbes's sovereign gazes out from above the composite body, crowned, bearing sword and crozier; Bear's collective has no crown, no sovereign, no representative person in whom its multitude is unified. It is an acephalous body politic, closer to the headless figure André Masson drew for Georges Bataille's review *Acéphale* (1936) than to Hobbes's sovereign. It is also, more subtly, a Pauline body in which every member's need of every other has become literal metabolic dependence.

The philosophical stakes of acephaly were being debated in Paris in the very year of the novelette. Jean-Luc Nancy's *La Communauté désœuvrée* (1983) argued that the dream of community as fusion, the immanent communion in which individuals dissolve into a single

body, is the essence of totalitarianism, and that genuine community consists rather in the sharing of finitude by singular beings who remain exposed to one another. Maurice Blanchot's *La Communauté inavouable*, published the same year in response, meditated on a community without unity or project. *Blood Music* poses their question in biological terms, and its answer is double. From outside, the noocyte collective is the nightmare of immanent fusion: bodies flow together into undifferentiated masses. From inside, it is closer to what Nancy would later call being singular plural (1996): persons persist as distinct singularities within the Thought Universe, multiplied rather than abolished. The discrepancy between the two descriptions is not a contradiction in the novel but the phenomenon that Chapter V formalises as the membrane theorem.

4.4 Continuity and the erotics of dissolution

The first symptoms of Vergil's transformation include an awakening of sexual vigour, and the agent passes from lover to lover: Edward discovers Vergil and Candice Rhine dissolving together into shapeless tissue, and Edward's infection passes to his wife Gail. The novel's erotics are not incidental. Georges Bataille's *L'Érotisme* (1957) begins from the proposition that human beings are discontinuous, each born and dying alone, and that eroticism, death and the sacred share a single movement toward the continuity from which individual existence separates us. Bataille illustrates the point with the reproduction of simple organisms: when a cell divides, there is a moment in which the one becomes two and a continuity passes between beings who will thereafter be discontinuous. The mitotic structure of *Blood Music*, from its cell-cycle titles to its final division of worlds, is Bataille's figure written into narrative form, and the lovers' dissolution is its erotic limit-case.

The literary genealogy of that limit-case is long. In Plato's *Symposium* (192d–e), Aristophanes imagines Hephaestus standing over two lovers and offering to melt and fuse them into one (*συντῆξαι καὶ συμφυσῆσαι εἰς τὸ αὐτό*), so that they would live as one being and, dying, die as one. In Wagner's *Tristan und Isolde* (1865), the lovers' Act II duet wishes for existence *ohne Nennen, ohne Trennen … endlos ewig ein-bewußt* ("without naming, without parting … endlessly, eternally, one-conscious"), and the opera ends in Isolde's ecstatic drowning in the world-breath. Bear's lovers achieve the Aristophanic fusion and the Wagnerian *ein-bewußt* literally, and the novel registers both the horror of the achievement, as Edward sees it from outside, and its consummation, as the converted experience it from within.

4.5 Inverse transubstantiation

The anthropology of religion since James Frazer's chapter on 'Eating the God' in *The Golden Bough* has recognised the theophagic logic of communion: the worshipper incorporates the god and is incorporated by him. Augustine heard that logic reversed in the voice that addressed him in the *Confessions*: he would not change the divine food into himself, as with

bodily food, but would himself be changed into it. The noocyte conversion is Augustine's inversion made biological. The converted do not digest the agent; the agent digests them, and they are changed into it.

The Thomistic doctrine of transubstantiation (*Summa Theologiae* III, q. 75) holds that in the Eucharist the substance of bread becomes the substance of Christ's body while the accidents (appearance, taste, texture) remain. The noocyte conversion is its exact inverse. The accidents of the human body (its shape, its organs, its visible individuality) perish entirely, while what made the person that person (memory, character, the pattern of a life) is preserved and incorporated into a larger body. One might call this *inverse transubstantiation*: the appearance goes, the essence remains, and the essence is assumed into a mystical body.

4.6 Informational resurrection

The Thought Universe is the novel's eschatological centre, and its premise, that no one is lost because every pattern is preserved and may be re-instantiated, places *Blood Music* within a lineage of resurrection projects that runs from scripture through Russian Cosmism to the physics of the Omega Point. Paul proclaimed that *we shall not all sleep, but we shall all be changed, In a moment, in the twinkling of an eye* (1 Corinthians 15:51–52), and Ezekiel's valley of dry bones (Ezekiel 37) gave the hope of collective revival its most vivid image. The Zoroastrian *frašō.kərəti*, the 'making wonderful' of existence at the end of time, joins the raising of the dead to the renovation of the world. Ovid's Pythagoras, in the fifteenth book of the *Metamorphoses*, states the principle in a single line: *omnia mutantur, nihil interit* (XV.165), "all things change, nothing perishes."

The modern secular form of the hope begins with Nikolai Fyodorov's *Philosophy of the Common Task* (published posthumously, 1906–13), which declared the scientific resurrection of every ancestor to be humanity's supreme moral obligation, and passes through Teilhard's Omega Point to Frank Tipler's *The Physics of Immortality* (1994), in which a final cosmic computer emulates every possible human life. The noocytes perform Fyodorov's common task in a form Fyodorov could not have imagined: not by reassembling scattered atoms but by preserving and re-instantiating information. The closing vignette of the novel, an adolescent romance from Bernard's past re-enacted within the transfigured order, is the common task at its most intimate. It resurrects not a body but a moment, and permits a life to revisit what it once failed to complete.

Informational resurrection raises the problem of personal identity in its most acute form, and the problem was being restated, in the year before the novel, by Derek Parfit. *Reasons and Persons* (1984) argued through cases of teletransportation, branching and fission that personal identity is not what matters in survival; what matters is psychological continuity and connectedness, which may hold one-to-many. Parfit reported that accepting this view freed him from what had seemed a glass tunnel, at the end of which lay darkness. The Thought Universe is a Parfitian world in which fission is routine and copies of a person may coexist; its consolations are available only to those who accept that the question 'will it

really be me?' has, at bottom, no further answer beyond the continuity of the pattern. *Blood Music* and *Reasons and Persons* are twin documents of the mid-1980s reductionist self, the one argued by cases, the other by fable.

The Mahāyāna tradition offers the most rigorous image of the order the noocytes create. The *Avataṃsaka Sūtra* envisions innumerable world-systems in the pores of the Buddha's body; the *Vimalakīrtinirdeśa* places Mount Sumeru within a mustard seed; and the Huayan school's image of Indra's net, in which each jewel reflects every other, articulates a principle of mutual interpenetration in which the one contains the all. The Thought Universe, in which every member carries the patterns of all the others and nothing is lost, is Indra's net rendered in nucleic acid, and the world within the pores is no longer a vision but an anatomy.

CHAPTER

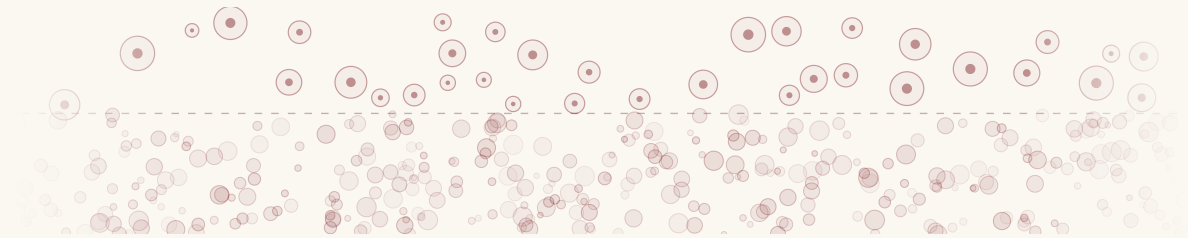
V

The Horror of the Plural

Blood Music and the Weird, the Protoplasmic, and the New Flesh

...as men busied themselves about their various concerns they were scrutinised and studied, perhaps almost as narrowly as a man with a microscope might scrutinise the transient creatures that swarm and multiply in a drop of water.

H. G. Wells, *The War of the Worlds* (1898), I.1



Wells opened the modern invasion narrative by placing humanity under the microscope: we are the transient creatures in the drop of water, studied by intelligences vaster than ours. *Blood Music* turns the instrument around. The creatures in the drop become the observers, and the observer at the eyepiece becomes the drop. This chapter situates that reversal within the history of horror, argues that the noocyte is the most fully interstitial monster the genre had produced, and formulates the principle, the membrane theorem, that explains why a book so saturated with horror is nonetheless experienced as a work of hope.

5.1 Theories of the monstrous

H. P. Lovecraft's essay 'Supernatural Horror in Literature' (1927) located the genre's source in the oldest and strongest of human emotions, fear, and in its oldest and strongest form, fear of the unknown. Noël Carroll's *The Philosophy of Horror* (1990), drawing on Mary Douglas's analysis of impurity as matter out of place (*Purity and Danger*, 1966), defined the monster of 'art-horror' as categorically interstitial: a being that is threatening because it is impure, and impure because it violates the categories by which a culture orders the world (living and dead, inside and outside, animal and human, one and many). Julia Kristeva's *Pouvoirs de l'horreur* (1980) described the abject as what disturbs identity, system and order, what does not respect borders, and found its paradigm in the corpse, the body that has become thing. Kelly Hurley's *The Gothic Body* (1996) borrowed William Hope Hodgson's word *abhuman* for the fin-de-siècle subject whose identity dissolves into morphic variability. Eugene Thacker has distinguished the world-for-us, the world-in-itself and the world-without-us, locating horror at the point where the second and third intrude upon the first; Mark Fisher distinguished the weird (that which does not belong) from the eerie (agency where none should be, or absence where agency should be).

By every one of these measures the noocyte is a limit-case. It is interstitial along at least six axes at once: self and other (it is the host's own cells), one and many (it is a person and a population), living and mechanical (it is a biological computer), part and whole (it is a constituent that becomes the totality), inside and outside (it originates within and emerges without), and cell and person (it is both a unit of tissue and a subject of discourse). It is abject in Kristeva's strict sense: its emergence turns the body into matter that has lost its form. It is weird, a presence that does not belong in the bloodstream, and eerie, an agency located where there should be none. It is the world-without-us manifesting from inside the world-for-us, a population of minds whose world is our flesh. And yet the novel concludes in beatitude. Carroll's model therefore needs supplementing. The interstitial is horrifying relative to a categorical scheme. When the scheme of the perceiver is itself rewritten, the interstitial ceases to be interstitial and becomes the new normal. The rest of this chapter traces that relativity through the genre's history.

5.2 The protoplasmic Gothic

Thomas Henry Huxley's lecture 'On the Physical Basis of Life' (1868) made *protoplasm* the common substance of all living things, and in the same year Huxley described, in mud dredged from the Atlantic floor, a primordial living slime he named *Bathybius haeckelii* in honour of Haeckel. The *Challenger* expedition showed in 1875 that *Bathybius* was a chemical precipitate, but for several years the Victorian imagination entertained the vision of the ocean floor carpeted with undifferentiated living matter, the primal ooze from which life arises. *Blood Music* realises Bathybius on dry land: a continent carpeted with protoplasm that is not primitive but supremely intelligent.

The fiction of the 1890s turned the protoplasmic vision into horror. In Arthur Machen's *The Great God Pan* (1894), the surgeon Raymond operates on a young woman's brain so that she may 'see the god Pan', and the daughter born of that vision, Helen Vaughan, dies in a scene of morphological dissolution in which her body passes through the stages of life backward, from human through animal to formless substance, and up again: the principle of life made visible as horror. In Machen's 'Novel of the White Powder' (1895), a student who takes a corrupted medicine is found reduced to a dark, putrid mass, neither liquid nor solid, still moving. William Hope Hodgson's 'The Voice in the Night' (1907) tells of a shipwrecked couple slowly overgrown and transformed by a grey fungus whose growth they come to crave. H. G. Wells contributed two decisive inversions. In *The Island of Doctor Moreau* (1896), the vivisected Beast Folk worship their maker as the lord of the House of Pain and recite his Law; they are creatures who take their creator for a god, as Bear's noocytes take their host. In *The War of the Worlds* (1898), the invaders are destroyed by terrestrial bacteria; Wells's microbes are humanity's saviours, and Bear's cells, a century of biology later, are its successors.

5.3 Lovecraft's servants and the Innsmouth turn

Lovecraft's *At the Mountains of Madness* (written 1931, published 1936) supplies the noocyte's most exact horror-ancestor. The shoggoths were engineered by the Elder Things as servant protoplasm, plastic tissue that could be shaped into whatever organ a task required; over aeons they acquired intelligence, imitated their masters, rebelled and overthrew them. The shoggoth is the noocyte's dark twin: biotechnology as slave tissue, the servant that becomes master. 'The Colour Out of Space' (1927) adds the motif of contamination that transforms an entire ecosystem, plants, animals and people, into something alien and luminous.

The Lovecraft text that matters most for the structure of *Blood Music*, however, is 'The Shadow over Innsmouth' (1936). Its narrator flees the half-human Deep Ones of a decaying Massachusetts port and then discovers, through genealogical research, that he is descended from them and has begun to change. In the final pages his horror inverts into anticipation: he resolves to go down to the sunken city and join his kin in a life of wonder beneath the sea. This *Innsmouth turn*, in which the affect of transformation reverses at the moment the narrating consciousness crosses from outside to inside, is the generic hinge on which Bear's

novel repeatedly swings. Edward Milligan crosses it at the end of the novelette, Bernard crosses it in Wiesbaden, and Suzy McKenzie, the novel's last unconverted witness, stands before it.

The contrast with Lovecraft is equally instructive. Lovecraft's 'cosmicism' turns on the indifference of the universe: the horror of being of no account to vast powers. Bear's horror turns on the opposite: the horror of total attention. The noocytes know their host molecule by molecule; they will remember every person they absorb. *O LORD, thou hast searched me, and known me* (Psalm 139:1) is the psalmist's comfort and, for the unconverted, the novel's terror. Lovecraft feared being unknown by the cosmos; Bear imagines being known completely by one's own flesh.

5.4 Assimilation and the homophonic collective

The mid-twentieth century produced a genre of assimilation narratives whose political allegory has been endlessly debated. In John W. Campbell's 'Who Goes There?' (1938) an Antarctic research party confronts an organism that absorbs and perfectly imitates living things. Its most celebrated scene is the blood test: because every fragment of the creature is a complete organism with its own will to survive, a sample of its blood recoils from a heated wire, betraying the impostor. Campbell had already imagined blood as an agent, each drop a self, and John Carpenter's film *The Thing* (1982), released the year before the novelette, made that scene a set piece of the decade's body horror. Robert Heinlein's *The Puppet Masters* (1951) imagined hive-communicating parasites riding human spines; Jack Finney's *The Body Snatchers* (1955), with its films of 1956 and 1978, imagined replicas grown in pods that are perfect in appearance and empty of feeling. John Wyndham's *The Midwich Cuckoos* (1957) imagined children who share a single mind. Star Trek's Borg, introduced in 1989, later condensed the tradition into a slogan of futile resistance.

These collectives are *homophonic*: the assimilated lose affect, memory, preference and voice, and the collective speaks with one. Their politics were read in the Cold War as images of communism or of conformist consumer society, but the formal feature is constant: assimilation is subtraction. Bear's collective is the opposite. The converted gain capacities; they persist as persons; they may be multiplied rather than erased; they converse, remember, revisit. The noocyte collective is *polyphonic*. Chapter VIII develops this distinction into the second principle of haematic polyphony. Here it suffices to note that *Blood Music* takes the genre's master fear, the loss of the self to the collective, and replaces it with a more disquieting proposition: that the collective may preserve the self better than the self could.

5.5 The New Flesh: Cronenberg, Barker, Butler, Otomo

Blood Music belongs to the most productive decade of body horror in the history of the form, and its contemporaries illuminate it. In David Cronenberg's *Shivers* (1975), a physician named, pointedly, Emil Hobbes engineers a parasite to return an over-rational humanity to the body, and it spreads through a modernist apartment block as erotic

contagion: a Leviathan inverted, the social contract dissolved by a designed organism. *Rabid* (1977), *The Brood* (1979), *Videodrome* (1983) and *The Fly* (1986) elaborated the 'new flesh', a mutation that is at once disease and transcendence, and whose ambiguity Cronenberg refused to resolve.

Clive Barker's *Books of Blood* (1984–85) produced two stories that read as glosses on Bear. In 'In the Hills, the Cities', two Yugoslav towns periodically construct colossal giants from their own strapped-together citizens to wrestle in the mountains; when one giant collapses, the other, maddened, walks off into the hills: Hobbes's frontispiece made flesh and set loose. In 'The Body Politic', a man's hands revolt against him, sever themselves and raise an insurrection of hands: the fable of Menenius Agrippa (Livy, *Ab Urbe Condita* II.32), in which the members rebel against the belly, told as horror in the same year as Bear's novel. Octavia E. Butler's *Clay's Ark* (1984) imagined an extraterrestrial micro-organism that compels its hosts to spread it and transforms their children into a new species, and her story 'Bloodchild' (1984) made the consent of a host body the explicit centre of a narrative of symbiosis; her *Xenogenesis* trilogy (1987–89) carried the problem to the genetic merger of species. Katsuhiro Otomo's *Akira* (manga 1982–90; film 1988) ends with an uncontrollable proliferation of flesh that swallows a city and opens onto the birth of a new universe. Rudy Rucker's *Software* (1982) and *Wetware* (1988) gave the cyberpunk decade its vocabulary of minds migrating across substrates.

Bear's novel shares the decade's conviction that the next transformation of the human will be somatic and that it will be both horror and grace. What distinguishes it is that it alone gives the new flesh a cosmology: an account of what the transformation means for the structure of physical reality and for the fate of every person it touches.

5.6 The inheritors

The structural problem Bear posed has been taken up repeatedly. Hideaki Anno's *Neon Genesis Evangelion* (1995–96) and *The End of Evangelion* (1997) culminate in the 'Human Instrumentality Project', in which the barriers that separate persons, figured as the 'AT Field' or Absolute Terror Field, are dissolved and humanity merges into a primordial fluid; the protagonist ultimately chooses the return of separate existence, with its pain. Instrumentality is homophonic where Bear's collective is polyphonic, but both narratives turn on consent, and both make the boundary of the self an explicitly emotional membrane. Jeff VanderMeer's *Annihilation* (2014) and the film of 2018 imagine a zone in which genetic identity is refracted and recombined, and treat transformation as a becoming the protagonist accepts. Peter Watts's 'The Things' (2010) retells Campbell's story from the perspective of the alien, for whom solitary, unshareable human individuality is the monstrosity and assimilation an act of communion. Watts's story is the clearest single-text demonstration of the principle stated below: the same event, narrated from the other side, changes affective sign. Against all of these, Peter Wessel Zapffe's essay 'Den sidste Messias' (1933) and Thomas Ligotti's *The Conspiracy Against the Human Race* (2010) represent the antithetical tradition, in which consciousness itself is the tragedy and its multiplication the

worst conceivable outcome. *Blood Music* is the great anti-Zapffean fiction: it treats the proliferation of consciousness as the remedy for the isolation that makes consciousness tragic.

5.7 The membrane theorem (first statement)

The evidence of this chapter supports a general proposition. Bear himself described the novel as both a frightening horror story and a tale of religious transcendence (Thacker 2001). That double character is not a mixture of tones but a function of position. A transformation that dissolves the boundary of the self is experienced as horror from the side of the boundary that has not yet been dissolved, and as rapture from the side that has. Horror and mythic beatitude are not genres but phase-states of a single narrative matter, separated by a membrane, and the affective valence of any transformation narrative is a function of the side of that membrane from which it is narrated. Table 2 sets out the corollary in the corpus surveyed.

Table 2. Locus of narration and affective valence in transformation narratives

Work	Locus of narration	Crossing of the membrane	Resulting affect
Finney, *The Body Snatchers* (1955)	Outside	Resisted throughout	Horror
Lovecraft, 'The Shadow over Innsmouth' (1936)	Outside, then inside	At the close	Horror inverted into homecoming
Clarke, *Childhood's End* (1953)	Outside (adults and Overlords)	Barred to the narrators	Elegiac sublime
Anno, *The End of Evangelion* (1997)	Inside, then outside	Crossed and refused	Ambivalence; separation chosen
Watts, 'The Things' (2010)	Inside (the collective)	Already crossed	Pity for the individual; communion
Bear, *Blood Music* (1985)	Alternating (Edward, Bernard, Suzy, the noocytes)	Repeatedly, by different subjects	Horror and beatitude in oscillation

The theorem predicts that narratives which remain outside the membrane are horror; narratives which remain inside are myth, utopia or rapture; and narratives which cross it, once or repeatedly, generate the characteristic oscillation of *Blood Music*. It also explains a feature of the novel that critics have found tonally puzzling: its apparent optimism in recounting the end of humanity. The optimism is not a lapse of tone. It is the view from inside.

CHAPTER

VI

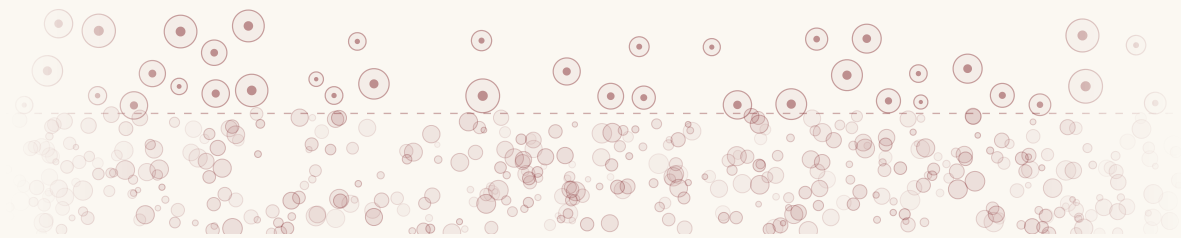
Mythopoetic Inversions

Dismemberment, Deluge, Metamorphosis, Ascension

sahasraśīrṣā puruṣaḥ sahasrākṣaḥ sahasrapāt

"Thousand-headed is the Person, thousand-eyed, thousand-footed."

Ṛgveda X.90.1



Lévi-Strauss proposed that a myth is the set of all its variants and that myths think themselves in men, generating new versions by systematic transformation of the old. On that view *Blood Music* is not merely a novel that alludes to myths but a new member of a transformation group whose oldest members are the cosmogonies of dismemberment. This chapter identifies the group, specifies the transformations Bear performs on it, and then sets the novel beside the mythopoetic fiction of the twentieth century.

6.1 Dismemberment and somatogony

Across Indo-European and East Asian traditions the world is made from a body. In the *Puruṣa-sūkta* (*Ṛgveda* X.90) the gods sacrifice the thousand-headed, thousand-eyed cosmic Person; from his mind the moon is born, from his eye the sun, from his breath the wind, from his parts the regions of the world and the orders of society; and the hymn declares *puruṣa evedaṃ sarvam*, the Person is all this (X.90.2). In the Old Norse *Grímnismál* (st. 40), *Ór Ymis holdi / var jǫrð of skǫpuð, / en ór sveita sær*: from Ymir's flesh the earth was shaped, and from his blood the sea, as mountains from his bones and sky from his skull. In the Chinese account preserved from Xu Zheng's *Wuyun linian ji* (third century CE), the dying Pangu becomes the world: his breath wind and cloud, his eyes sun and moon, his blood the rivers, his flesh the fields. In the Babylonian *Enūma eliš* (Tablet IV), Marduk splits the body of Tiamat to make heaven and earth.

Blood Music runs this operation backward. The world is not carved from a body; a world is re-membered into one. Human beings, cities, forests and rivers are gathered into a continental organism that is, in the most literal sense the genre has achieved, a cosmic Person in the process of assembly. This inversion of cosmogony may be named *somatogony*: the generation of a body from a world. The pun latent in English is exact. Where the ancient cosmogonies dismember, Bear re-members, and the Thought Universe, in which every absorbed life is retained, is a re-membering in the anamnestic sense as well: an act of total recollection.

Two details of the ancient myths reveal that the transformation group already contained the noocyte in embryo. Snorri Sturluson's *Gylfaginning* relates that the dwarves first took shape and quickened in Ymir's flesh as maggots, and that by decree of the gods they acquired human understanding and the likeness of men. And in the account of Pangu, the parasites on the giant's body, touched by the wind, were transformed into the black-haired people, humanity itself. Two independent mythologies thus hold that conscious beings arose as vermin in the flesh of a cosmic body and were raised to understanding. Bear's noocytes recapitulate the motif with exactness: consciousness kindled in the flesh of a being who is, to them, the cosmos. The difference is the direction of the sequel. Ymir's maggots become dwarves who live in the world made from his corpse; Pangu's parasites become people who live on the world made of his body. Bear's cells, raised to understanding, take the cosmic body back.

The Upanishadic tradition supplies the cosmogony of division that the novel's form reproduces. In the *Bṛhadāraṇyaka Upaniṣad* (I.4.1–3), in the beginning there is only the Self in the form of a person; finding no delight in being alone, it desires a second, and causes itself to fall into two. Cosmogony is a cell division motivated by loneliness. *Blood Music* answers with a double movement: inward fusion and outward fission. The many reunite within the collective, undoing the primal loneliness, while the universe itself divides, at the end, into two daughter worlds.

6.2 Blood, intelligence, and the drum

Mesopotamian anthropogony brings the motif still closer to the title. In *Enūma eliš* (Tablet VI), humankind is fashioned from the blood of Kingu, the rebel god, to bear the labour of the gods. In the Old Babylonian *Atrahasis* (Tablet I), the gods slaughter a deity who possessed *ṭēmu*, intelligence or planning, and the birth-goddess mixes his flesh and blood with clay to make the first humans; thereafter, the text says, the drum is heard for all time, conventionally understood as the human heartbeat, and a ghost (*eṭemmu*) of the slain god abides in human flesh. Here, in the oldest surviving anthropogony of the Near East, intelligence is located in blood, and its presence in the human body is audible as a rhythm. The *Atrahasis* drum is the first blood music.

The same epic supplies, two thousand years before Bear, the logic of the Ulam threshold. Human beings multiply, and their noise (*rigmu*, *hubūru*) grows until the land bellows like a bull and Enlil cannot sleep. The gods respond with plague, drought, famine and, finally, the flood. The threat to cosmic order in *Atrahasis* is precisely the density of a multiplying population whose collective voice the divine order cannot bear. In Bear, a multiplying population of minds raises a din of observation that physical law cannot bear. The resolution differs, and the difference is the novel's ethical signature. In *Atrahasis* the gods destroy the noisy population. In *Blood Music* the noisy population withdraws.

6.3 Deluge and the softening of stone

The outbreak travels by water (Vergil's contemplated release into the municipal supply is its pivot), and the novel's catastrophe has the shape of a deluge: a continent submerged not by water but by the tide of a new kind of life, while the rest of the world watches from the far shore of a quarantine. Plato's Atlantis, an island continent swallowed in a single day and night (*Timaeus* 25c–d), and the floods of *Gilgamesh* XI and Genesis 6–9 supply the ancient archive. Genesis 1:28 supplies the imperative that the noocytes obey more perfectly than any creature before them: *Be fruitful, and multiply, and replenish the earth.*

Ovid's account of the aftermath of Deucalion's flood (*Metamorphoses* I.313–415) offers the most precise inverse. Instructed to cast behind them *ossa magnae parentis*, the bones of the great mother, Deucalion and Pyrrha throw stones over their shoulders, and the stones soften and take human form: what was earthy and moist becomes flesh, what was solid becomes bone, and *quae modo vena fuit, sub eodem nomine mansit*, "what was just now a vein

remained under the same name" (I.410). After Ovid's flood, earth becomes flesh. After Bear's, flesh becomes landscape: human bodies soften, flow and settle into the living terrain of the converted continent. And the veins remain veins.

6.4 Metamorphosis against Kafka

Ovid's poem is a theology of continuous change, culminating in the Pythagorean discourse of Book XV and its principle that all things change and nothing perishes. Apuleius's *Metamorphoses*, the *Golden Ass*, adds the initiatory model: Lucius's transformation into an ass and his restoration by Isis are stages of a mystery whose end is the soul's illumination (Book XI). Franz Kafka's *Die Verwandlung* (1915) gave the twentieth century its counter-myth: Gregor Samsa's transformation isolates him from his family, who are relieved by his death. *Blood Music* is anti-Kafkan in exact measure. Its transformations are initiatory rather than isolating; they reunite rather than separate. Suzy McKenzie's family is absorbed while she remains outside, and the novel's question about her is never whether she will be rejected for having changed, but whether she will choose to join those who have.

6.5 The consensual rapture

The disappearance of the converted from the physical world, and the persistence of the unconverted on an emptied earth, structurally recalls the dispensationalist doctrine of the rapture, formulated by John Nelson Darby in the 1830s on the basis of 1 Thessalonians 4:17 (*Then we which are alive and remain shall be caught up together with them in the clouds*) and popularised in the decade before the novel by Hal Lindsey's best-selling *The Late Great Planet Earth* (1970). Bear secularises the rapture and reverses its moral valence. In the dispensationalist scheme, those left behind are the unsaved, condemned to tribulation. In *Blood Music*, those who remain do so because their refusal has been respected. Being left behind is not a sentence but a choice. The Norse eschatology supplies the coda: after Ragnarök, the seeress of the *Vǫluspá* sees the earth rise a second time from the sea, ever green (*Sér hon upp koma / øðru sinni / jǫrð ór ægi / iðjagræna*, st. 59). The final Interphase of Bear's novel is that second rising: a physical world restored to those who chose it.

6.6 Mythopoetic science fiction

The twentieth century produced a body of fiction whose ambition was explicitly mythological: to supply, for a scientific cosmology, the narratives that religion had supplied for older ones. *Blood Music* belongs to this tradition, and its place within it can be fixed by four lineages.

The microcosmic lineage. Fitz-James O'Brien's 'The Diamond Lens' (1858) tells of a microscopist who falls in love with a woman glimpsed in a drop of water and loses her when the drop evaporates. Ray Cummings's *The Girl in the Golden Atom* (1919–22) populated the atom with civilisations. Theodore Sturgeon's 'Microcosmic God' (1941) is the most direct generic ancestor of the novel: a reclusive scientist breeds a race of tiny, rapidly

evolving beings, the Neoterics, who worship him as god and solve his problems, and who finally seal themselves behind an impenetrable shield. James Blish's 'Surface Tension' (1952) imagined engineered microscopic humans in an alien pond, building a 'spaceship' to cross to the next puddle. Commentators have placed *Blood Music* alongside Clarke, Sturgeon and Blish (eNotes), and the lineage clarifies Bear's innovation. Sturgeon's microcosmic god remains outside his creatures' world, a benevolent tyrant. Bear's microcosmic god is the world, and his creatures live inside him.

The Stapledonian lineage. Olaf Stapledon's *Last and First Men* (1930) imagined Martian invaders composed of sub-vital units that combine into cloud-like group minds, and later human species that achieve racial telepathic unity. *Star Maker* (1937) traced the ascent of mind from individual creatures through symbiotic races and galactic communities to a cosmic spirit that at last glimpses its maker, an artist who regards his creations with an unsettling, contemplative detachment. Critics have repeatedly placed Bear among the few writers capable of following where Stapledon led. *Blood Music* compresses Stapledon's billion-year ascent into a single season and locates it in a single bloodstream.

The Clarkean lineage. David Langford's comparison with *Childhood's End* (1953) is just: there too a generation of humanity merges into a collective mind (the Overmind), the earth is consumed in the ascent, and a remnant watches. Clarke's Overlords conceive of themselves as midwives to a transformation they cannot share, and Bear's Edward Milligan, an obstetrician, plays that role in miniature, attending at a birth that requires his friend's death. The decisive difference is consent. Clarke's adults are excluded by nature from the transfiguration of their children; Bear's unconverted are excluded only by their own refusal, and the refusal is honoured. *Childhood's End* is an elegy for those who cannot cross; *Blood Music* is a meditation on those who need not.

The Lemian lineage. Stanisław Lem's *Solaris* (1961) imagined a planetary ocean whose intelligence defeats human understanding; *The Invincible* (1964) imagined a swarm of micro-machines whose collective behaviour constitutes an intelligence without a mind; and *Golem XIV* (1981) imagined a superintelligent computer that lectures its makers on the limits of their nature and then falls silent, having passed beyond the threshold of communication. Golem XIV's withdrawal is the closest antecedent to the noocytes' exodus, and Lem's *Summa Technologiae* (1964) had already speculated on the breeding of information and the evolution of intelligence beyond its biological origins. Later, Vernor Vinge's Transcend (*A Fire Upon the Deep*, 1992), Iain M. Banks's Sublime (*The Hydrogen Sonata*, 2012) and Greg Egan's *Diaspora* (1997) would elaborate the departure of minds from the physical universe into a full cosmological genre.

Two further works define the novel by contrast. C. S. Lewis's *That Hideous Strength* (1945) imagined a technocratic conspiracy, guided by 'macrobes', whose ideal is mind freed from organic life, a clean world without the mess of biology. Bear's noocytes are that dream's antithesis: mind that saturates organic life rather than escaping it. Jorge Luis Borges's 'Tlön, Uqbar, Orbis Tertius' (1940) imagined an idealist world in which objects are

duplicated by expectation and fade when forgotten, and which gradually overwrites our own. *Blood Music* is Tlön in tissue: a perception-dependent reality advancing upon the shared one, until the shared one must be partitioned.

6.7 The transformation group

Table 3 sets out the transformation group reconstructed in this chapter, specifying for each myth the body involved, the operation performed on it, the result, and the transformation *Blood Music* performs.

Table 3. Bear's transformations of cosmogonic and anthropogonic myth

Myth	Body	Operation	Result	Transformation in *Blood Music*
Puruṣa-sūkta (*Ṛgveda* X.90)	Thousand-eyed cosmic Person	Sacrificial dismemberment	World and social order	World re-membered into a many-eyed Person
Ymir (*Grímnismál*; *Gylfaginning*)	Primordial giant	Dismemberment by the gods	Earth, sea of blood, sky; maggots in the flesh raised to understanding	Cells in the flesh raised to understanding reclaim the body
Pangu (*Wuyun linian ji*)	Dying primordial giant	Transformation of parts	Rivers of blood; parasites become humanity	Constituents become humanity's successors
Tiamat and Kingu (*Enūma eliš* IV, VI)	Primordial sea; rebel god	Splitting; bloodletting	Heaven and earth; humans from rebel blood	A new mind from rebellious blood
Atrahasis I	God possessing intelligence	Flesh and blood mixed with clay	Humans with a heartbeat 'drum'; their multiplying noise brings the flood	Intelligence in blood; noetic density answered by withdrawal, not destruction
Deucalion and Pyrrha (Ovid, *Met.* I)	The great mother's bones	Stones thrown behind	Stone softened into flesh; veins remain veins	Flesh softened into living landscape
Bṛhadāraṇyaka Upaniṣad I.4	The lonely primordial Self	Division into two	A world of pairs	Inward fusion of the many; outward fission of the world

CHAPTER

VII

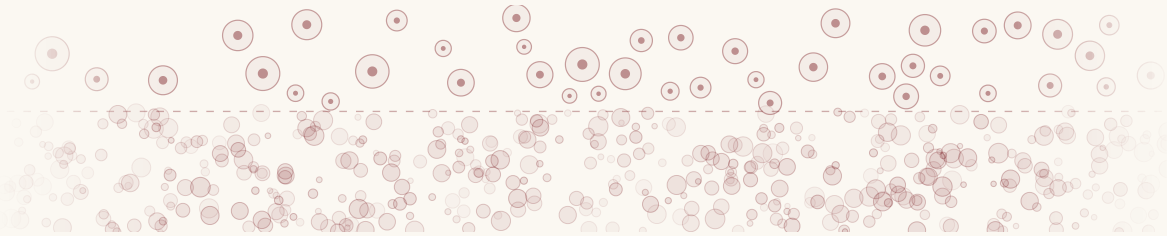
The Dantean Armature

Vergil, Bernard, and the *Commedia* of the Cell

Sangue perfetto, che poi non si beve / da l'assetate vene, e si rimane / quasi alimento che di mensa leve, / prende nel core a tutte membra umane / virtute informativa.

"Perfect blood, which is never drunk by the thirsty veins but remains behind like food one clears from the table, takes in the heart an informing power for all the human members."

Dante, *Purgatorio* XXV.37–41



7.1 The problem of names

Names in fiction are evidence of a peculiar kind. A single resonant name proves nothing; authors name characters for sound, for private reasons, or by accident, and the intentional fallacy, as W. K. Wimsatt and Monroe Beardsley formulated it in 1946, warns against treating an author's presumed design as the measure of meaning. Umberto Eco's distinction between the intention of the author and the intention of the work (*intentio operis*) offers a better criterion: an interpretation is warranted when it is confirmed by other parts of the text and does not violate them, and onomastic readings become compelling not through any single name but through convergence, when several independent features point in the same direction. This chapter argues that the names of *Blood Music* converge, and that they converge on Dante.

7.2 Vergil, the guide who may not enter

The protagonist's given name is spelled in the scholarly form of the Roman poet's name, Publius Vergilius Maro: *Vergil*, not *Virgil*. In the *Commedia*, Virgil is the pilgrim's first guide, through Hell and up the mountain of Purgatory, and he is barred from Paradise. At the summit, as Beatrice appears, he vanishes (*Purgatorio* XXX), returning to Limbo with the virtuous pagans who lived before grace. Vergil Ulam occupies the same structural position. He opens the way, describing his own experiment as a doorway through which life may pass beyond the human species, and he is removed, by his friend's hand, before the transfiguration he has made possible. Like his namesake he leads others toward a beatitude he is not permitted to inhabit in his own person.

The historical Vergil supplies a second correspondence that is, if anything, more precise. In the sixth book of the *Aeneid*, Anchises explains to his son the doctrine of the world-soul: *Principio caelum ac terras camposque liquentis / lucentemque globum Lunae Titaniaque astra / spiritus intus alit, totamque infusa per artus / mens agitat molem et magno se corpore miscet* (VI.724–727). "From the beginning, a spirit within sustains the heavens and the lands and the watery plains, the shining globe of the Moon and the Titan stars; and mind, poured through the limbs, moves the whole mass and mingles itself with the great body." No single sentence in classical literature comes closer to describing the noocyte continent: *mens*, infused *per artus*, through the joints and members, mingling *magno corpore*, with a great body. Anchises goes on to explain that purified souls drink from Lethe so that, forgetting, they may return to bodies (VI.748–751). The Thought Universe is the anti-Lethe: a return without forgetting, in which nothing is lost.

7.3 'I am Gulliver'

The protagonist's full name, Vergil I. Ulam, is a perfect anagram of *I am Gulliver*, a fact noted in critical commentary as signalling the novel's preoccupation with the minuscule (eNotes). The eleven letters correspond exactly, and the middle initial exists, it would seem, to make the anagram work. Jonathan Swift's *Gulliver's Travels* (1726) is the founding satire of relative scale. In Lilliput, Gulliver's body is a landscape to its tiny inhabitants, who bind

the sleeping giant with innumerable threads. In Brobdingnag, magnified human skin becomes monstrous, and Gulliver sees the vermin on a beggar's clothes more distinctly with the naked eye than he had ever seen a European louse through a microscope (Part II, ch. 4). Vergil is a Gulliver among his own Lilliputians, the giant whose body is their country and who is bound by them at last. Swift's Academy of Lagado, with its projectors extracting sunbeams from cucumbers, supplies the satirical type of the experimental scientist that Vergil, with his falsified credentials and private projects in a corporate laboratory, updates.

7.4 Ulam: automaton, chain reaction, singularity

The surname completes the onomastic overdetermination. Stanislaw Ulam (1909–1984), who died between the publication of the novelette and the novel, is associated with three ideas that *Blood Music* fuses. First, the cellular automaton: it was Ulam who suggested to von Neumann the lattice of discrete cells on which self-reproducing automata could be modelled. Second, the chain reaction: Ulam devised, with von Neumann and Nicholas Metropolis, the Monte Carlo method in order to calculate neutron multiplication in fissile material, and co-authored with Edward Teller the staged design of the thermonuclear weapon (1951). Third, the singularity: in his memorial tribute to von Neumann (1958), Ulam recorded a conversation in which von Neumann spoke of the ever-accelerating progress of technology as approaching an essential singularity in human history, beyond which human affairs as known could not continue. This is the earliest recorded use of 'singularity' in its now-familiar technological sense. Whether Bear intended the allusion is unknown. The convergence is exact nonetheless: the surname of the man who carries self-replicating cells into the world names the inventor of cellular self-replication, the theorist of the chain reaction, and the first witness of the singularity.

7.5 Bernard, the mystic of melting

The novel's second guide into the interior world bears the name of the last guide of the *Commedia*. In *Paradiso* XXXI–XXXIII, Bernard of Clairvaux replaces Beatrice, explains the order of the Celestial Rose, prays to the Virgin on the pilgrim's behalf, and conducts him to the beatific vision. Michael Bernard is the character who, in quarantine at Wiesbaden, converses with the cells within him, learns the physics of the new order, consents to his transformation, enters the interior world and passes into the Thought Universe, where he is honoured as a creator. The structural homology (first guide removed at the threshold, last guide conducting the passage into the final vision) is complete.

The historical Bernard deepens the correspondence. In *De diligendo Deo* (c. 1128), Bernard distinguishes four degrees of love, the highest of which is a union so complete that the self seems to dissolve. He illustrates it with three images: a small drop of water poured into much wine, which seems to lose itself entirely as it takes on the wine's taste and colour; iron in fire, which becomes like fire; and air flooded with sunlight, which seems to become light itself. In that state, he writes, every human affection must in some ineffable way *a semetipsa liquescere*, melt away from itself, and pour wholly into the will of God; yet *manebit*

quidem substantia, sed in alia forma, alia gloria aliaque potentia, "the substance will remain, but in another form, another glory, another power" (X.27–28). Bernard's doctrine of deifying love is the theological template of inverse transubstantiation (Chapter IV): the person melts, the substance remains, the form is new. That Bear's Bernard is the character who consents to melt into the collective and who finds, within it, that no one is lost, is the most striking of the novel's convergences.

The forename adds a further register. Michael is the archangel of the war in heaven (Revelation 12:7) and, in Christian tradition, the psychopomp who weighs and conducts souls. And the name Bernard bears one more association with the microcosm: the twelfth-century *Cosmographia* of Bernardus Silvestris, which divides creation into *Megacosmus* and *Microcosmus*, the great world and the small, was among the most influential medieval elaborations of the doctrine the novel literalises.

7.6 Purgatorio XXV: the canto of informing blood

The hinge of the Dantean reading is a canto that has nothing to do with names. In *Purgatorio* XXV, as the pilgrim climbs from the terrace of gluttony, he asks how the bodiless shades can grow thin. Statius answers with a discourse on embryology, the most sustained passage of natural philosophy in the poem, built on Aristotle's *De generatione animalium* and its medieval commentators. Its sequence is, point by point, the sequence of *Blood Music*.

First, blood carries form. The 'perfect blood' that the veins never drink acquires in the heart an informing power (*virtute informativa*) for all the members of the human body, as the blood that flows through the veins becomes those members (XXV.37–42). Bear's premise is the same proposition restated in the vocabulary of molecular biology: blood carries the information that builds the body, and the etymology of 'information', from *informare*, to give form, makes the identity of the two doctrines a matter of strict philology.

Second, the soul develops through stages. The active power becomes a soul like that of a plant, then moves and feels *come spungo marino*, like a sea-sponge (XXV.56), then develops organs for the powers of which it is the seed. The noocytes likewise ascend from cellular computation through animal-level intelligence toward something beyond the human.

Third, the rational soul is reflexive. When the brain is complete, God breathes into it a new spirit that draws the active elements into its own substance and makes *un'alma sola, / che vive e sente e sé in sé rigira* (XXV.74–75), "a single soul that lives and feels and circles back upon itself." The phrase *sé in sé rigira* is Dante's anticipation of what Hofstadter would call a strange loop: consciousness as self-reference. The noocytes become conscious exactly when they become able to model themselves and their host.

Fourth, and most remarkably, the soul builds a new body after death from a new medium. When the soul falls to its appointed shore, *la virtù formativa raggia intorno / così e quanto ne le membra vive* (XXV.89–90), "the formative power radiates around, in the same manner and as much as in the living limbs," and impresses upon the surrounding air a shape, the shade, which serves it as a body and through which it speaks, laughs, weeps and sighs. Dante's aerial body is a medieval theory of substrate-independent personhood: the

formative power of a person can organise whatever medium surrounds it. The Thought Universe, in which persons are re-instantiated in a substrate that is no longer flesh, is the *virtù formativa* radiating into a new air.

7.7 The commedia of the cell

With these correspondences established, the novel's large-scale form can be read as a commedia in three movements.

Its *Inferno* is the body horror of the opening movements: Vergil's descent into his own interior, the dissolution of the lovers, the death of the guide. The dissolution of Vergil and Candice into a single shapeless mass has a precise infernal antecedent. In *Inferno* XXV, the thieves are punished by the theft of their own forms: a serpent and a man fuse until *ogne primaio aspetto ivi era casso: / due e nessun l'imagine perversa / parea* (XXV.76–78), "every former aspect was cancelled there; the perverse image seemed both two and none." Dante pauses to bid Lucan and Ovid fall silent, claiming to have surpassed them as poets of metamorphosis (XXV.94–102). Vergil Ulam is a thief, of corporate property, of credentials, of the fire of a new kind of life, and the fusion of the lovers is the thieves' *contrapasso*: those who appropriated what was not theirs lose even the last property, their own form.

Its *Purgatorio* is the long quarantine at Wiesbaden: suffering endured in expectation of transformation, under the gaze of a hostile crowd, during which Bernard learns the theory of the new order much as the pilgrim learns from Statius the theory of informing blood.

Its *Paradiso* is the Thought Universe. Dante's Celestial Rose (*Paradiso* XXX–XXXII), in which the blessed sit in distinct and named places within a single flower, is the medieval image of a polyphonic collective: unity without fusion, each soul individuated within the one bloom. In the final canto Bernard's prayer addresses the Virgin as *figlia del tuo figlio* (XXXIII.1), daughter of her own son, the creature who bore her creator; the noocytes, daughters of their host, become the matrix in which the host is reborn. The pilgrim then sees in the depth of the divine light *legato con amore in un volume, / ciò che per l'universo si squaderna* (XXXIII.86–87), "bound by love in a single volume, all that is scattered in leaves throughout the universe." That is the Thought Universe's governing principle, the gathering of every dispersed life into a single book in which nothing is lost, stated in the language of the book of life that molecular biology would make literal. Even the novel's ending has its Dantean warrant. Bear closes on a dreamlike re-enactment of Bernard's first romance; the whole *Commedia* is set in motion by Dante's first love, recorded in the *Vita Nuova*. The Thought Universe offers, literally, a new life in which first love is revisited.

7.8 Convergence

The argument does not require that Bear planned a Dantean allegory. It requires only that the features enumerated (the first guide's name and fate, the last guide's name and mystical doctrine, the anagram of scale, the surname of the automaton and the chain reaction, the cell-cycle architecture, the obstetrician who attends the birth, and the thematic coincidence with Statius's embryology) belong to the semantic field of the work and

reinforce one another. Their convergence makes *Blood Music* legible as a scientific *Commedia*: a journey from the inferno of the body, through the purgatory of quarantine, to a paradise of total memory. It also discloses the medieval depth of the novel's governing idea. Long before molecular biology, Christian Aristotelianism held that blood carries the power to give form, that the soul is self-reflexive, and that personhood can outlive its flesh by impressing its form upon a new medium. Bear's achievement is to have restated that doctrine in the vocabulary of the cell and then asked what follows if the blood begins to inform not only its own body but the world.

CHAPTER

VIII

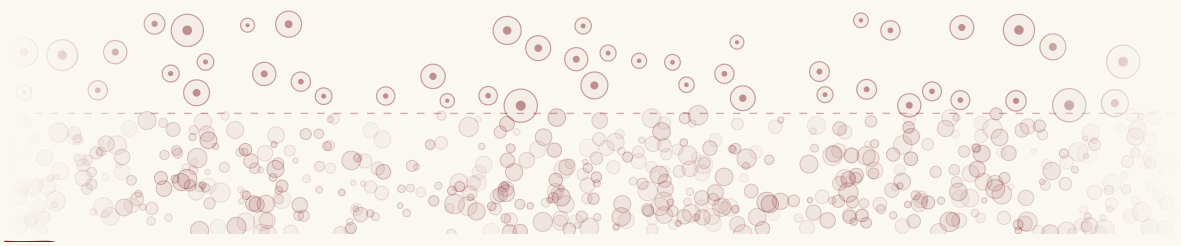
Haematic Polyphony

A Theory in Six Principles

Humanam vero musicam quisquis in sese ipsum descendit intellegit.

"Whoever descends into himself understands human music."

Boethius, *De institutione musica* I.2



Boethius divided music into three kinds: *musica mundana*, the harmony of the heavens and the elements; *musica humana*, the consonance that binds the incorporeal vitality of reason to the body, as low and high voices are tempered into a single concord; and *musica instrumentalis*, the music of lyre and pipe. He added that whoever descends into himself understands the second. *Blood Music* is the story of two men who descend into themselves and hear it, and of a music that, once heard, grows from *humana* into *mundana* and then passes beyond the world altogether. The theory proposed in this chapter takes its name from that trajectory. *Haematic polyphony* designates a condition of being, first narrated in Bear's novel, that can be stated in six principles. Two are ontological, one physical, one cosmogonic, one aesthetic and epistemic, and one eschatological and ethical. Each is derived from the novel and from the genealogies reconstructed in the preceding chapters, and each claims more than any of its sources.

8.1 First principle: informative virtue

Statement. In the world of *Blood Music*, form is not imposed on matter from a privileged locus (a soul, a nucleus, a code, a sovereign) but generated by computation distributed through matter itself. Blood informs itself.

The Western tradition, from Aristotle to molecular biology, placed the source of form above or outside the matter it shapes. In Aristotle's *De generatione animalium*, the male contributes form and the principle of movement while the female contributes matter, the menstrual blood (*katamēnia*) that the form informs; Statius's discourse in *Purgatorio* XXV transmits this gendered hylomorphism to Dante, and Plato's *Timaeus* had already figured the receptacle of becoming as a nurse (*τιθήνη*, 49a) who receives forms without possessing any. The twentieth century relocated the paternal form to the genome. Schrödinger's code-script and Francis Crick's 'Central Dogma' (1958), which held that sequence information flows from nucleic acid to protein and never back, made DNA the commanding text and the cell its executor. Evelyn Fox Keller's *Refiguring Life* (1995) and Lily Kay's *Who Wrote the Book of Life?* (2000) have shown how much of this picture was carried by metaphors of command, program and script.

Bear's noocytes overturn the hierarchy. Their form is not dictated by a master text but negotiated among agents; the 'program' rewrites itself; information is not instruction but conversation. The first breach in the Central Dogma, the discovery of reverse transcription by Howard Temin and David Baltimore in 1970, revealed that retroviruses write RNA back into DNA, and the retrovirus that would become the emblem of the decade in which Bear wrote is exactly such an agent. The noocytes generalise reverse transcription into a politics: the parts write back to the whole. Two feminist reconsiderations of the hylomorphic tradition sharpen the point. Judith Butler's *Bodies That Matter* (1993) exposed the gendered hierarchy in which form is masculine and matter feminine, and Julia Kristeva's *La Révolution du langage poétique* (1974) theorised a *semiotic chora*, borrowing Plato's term, as the rhythmic, musical, pre-linguistic space of drives associated with the maternal body, prior to and subversive of the symbolic order. Vergil first perceives his cells as music, then hears them speak, and finally passes, in Bernard, into an order beyond speech. The noocytes trace

the passage from the semiotic to the symbolic and beyond it, and that passage is witnessed by an obstetrician, a physician of the matrix. Blood, which Aristotle made the passive matter of generation, becomes in Bear the active principle of form.

8.2 Second principle: plural subjectivity

Statement. The subject is an unmerged choir of agents. Consciousness is not a monarch but a consonance; and a collective that preserves the independence of its voices is polyphonic, not homophonic.

Boethius described the union of mind and body as a tempering of low and high voices into one consonance. Mikhail Bakhtin, in his study of Dostoevsky (1929; revised 1963), took the term *polyphony* from music to describe a novel composed of a plurality of independent, unmerged voices and consciousnesses, each fully valid, none subordinated to the author's monologic perspective. The distinction between polyphony and homophony is the distinction drawn in Chapter V between Bear's collective and the assimilating collectives of the genre: the pods, the Borg, Instrumentality, in which incorporation is subtraction and the collective speaks with one voice.

Music theory supplies a precise formal criterion. In the discipline of species counterpoint codified by Johann Joseph Fux's *Gradus ad Parnassum* (1725), parallel perfect fifths and octaves are forbidden. The reason is acoustic and conceptual at once: voices that move in parallel perfect consonance fuse perceptually into one, and the independence of the parts, which is the whole point of polyphony, is lost. The central rule of contrapuntal composition is an anti-fusion rule. A polyphonic collective is one that observes it: its members may sound together, and must, but may not collapse into a single line. Dante's Celestial Rose, in which every blessed soul occupies a distinct and named place within a single flower, and Jean-Luc Nancy's later formula of being *singular plural*, are the theological and philosophical statements of the same rule. Bear's novel states it formally as well as thematically: its narrative is distributed among Vergil, Edward, Bernard, Suzy and the noocytes themselves, so that the book is a polyphony about polyphony.

The principle also resolves an antinomy in Nietzsche. *Die Geburt der Tragödie* (1872) opposed the Apollonian *principium individuationis*, Schopenhauer's term for the principle that divides the world into separate beings, to the Dionysian ecstasy in which, under the spell of music, the bond between human beings is renewed and estranged nature celebrates her reconciliation with her lost son. Nietzsche treated the two as complementary forces whose union produced tragedy, but in his account the Dionysian always threatens to annihilate the individual. *Blood Music* imagines what Nietzsche did not: a Dionysian dissolution of bodies that preserves the Apollonian integrity of persons. The flesh flows together; the voices remain distinct. Haematic polyphony is the name for that reconciliation.

8.3 Third principle: noetic density

Statement. The lawfulness of the physical world varies inversely with the concentration of its legislating observers. Beyond a critical density, the Ulam threshold, physical law ceases to be a shared consensus and becomes locally negotiable.

Chapter III formalised the principle and corrected it against quantum Darwinism: what destabilises a world is not the number of its witnesses but the number of its legislators, observers whose participatory choices are neither coordinated by a single will nor bounded by a common scale. The principle can now be generalised by analogy with tonality. Physical law, in this model, functions like a key: a shared tonal centre within which every voice, however independent, resolves. A tonal system can harmonise a great many voices, but not an unlimited number of voices each free to choose its own tonal centre. When the density of independent legislating voices exceeds what the system can reconcile, it must modulate. The noocytes' departure is a modulation into a key beyond the physical world, performed in such a way that the original key remains intact for those who stay. It is, in the strict sense, an enharmonic exit: a change of meaning without a change of sound for those who remain behind.

The principle's most consequential implication is political. If physical law is maintained by the sparseness of its legislators, then lawfulness is a commons: a finite, shared resource, depletable by overuse. Garrett Hardin's 'The Tragedy of the Commons' (1968) predicted that rational agents will exhaust any open-access resource; Elinor Ostrom's *Governing the Commons* (1990) showed that communities can and do govern shared resources through self-imposed rules. The noocytes confront a tragedy of the *nomic* commons, the commons of physical law itself, and they resolve it in Ostrom's manner, by collective self-limitation rather than by Hardin's predicted exhaustion.

8.4 Fourth principle: somatogony

Statement. Every cosmogony of dismemberment implies a possible somatogony of re-membering, in which a world is gathered into a body; and this gathering proceeds by a double movement, inward fusion and outward fission.

Chapter VI established that the myths of Puruṣa, Ymir, Pangu and Tiamat derive the world from a primordial body, and that two of them (Ymir's maggots raised to understanding, Pangu's parasites transformed into people) already contain the motif of conscious beings kindled in the cosmic flesh. *Blood Music* narrates the inverse operation as natural history. James Lovelock's Gaia (1979) had described the biosphere as a self-regulating system without attributing mind to it; Vernadsky had treated the noosphere as a geological force. Bear's continent is Gaia with a mind and the noosphere made flesh, and it suggests a name for the epoch it inaugurates, in which mind becomes first a geological and then a physical force: the *Noocene*. The double movement is essential to the principle. The many fuse inwardly, reversing the primordial loneliness of the *Bṛhadāraṇyaka* Self; the world divides outwardly, as the noetic and material orders separate. A somatogony re-members a body and, in the same act, divides a universe.

8.5 Fifth principle: the membrane theorem

Statement. The affective valence of a transformation that dissolves the boundary of a self is a function of the side of that boundary from which the transformation is narrated. Horror and mythic rapture are not genres but phase-states of one narrative matter.

Let T be a transformation that dissolves the boundary B of a self-system S, and let N be a narrating consciousness. The theorem may be written:

$$V(T \mid N) = \begin{cases} \text{horror} & \text{if } N \notin T(S) \\ \text{rapture} & \text{if } N \in T(S) \\ \text{oscillation} & \text{if } N \text{ crosses } B \end{cases}$$

Three corollaries follow. The first is generic: horror, myth and the hybrid form of *Blood Music* are distinguished not by subject matter but by the position of the narrating consciousness (Table 2). The second is psychoanalytic. Freud's essay 'Das Unheimliche' (1919) observed that the German *heimlich*, homely, develops in its range of meanings until it coincides with its opposite, *unheimlich*, uncanny. The noocyte is the uncanny in its purest form, the most intimate thing, one's own blood, become the most alien; and the crossing of the membrane runs Freud's semantic development in reverse, as the uncanny becomes home again. The third corollary is epistemic and ethical, and it is the theorem's most important consequence. Judgements about transformative change are subject to a systematic *membrane bias*: those outside a transformation will tend to perceive horror, and those inside it rapture, and neither perception is privileged by its position alone. L. A. Paul's *Transformative Experience* (2014) has shown that some choices cannot be rationally evaluated in advance, because the experience they lead to would change the very preferences by which one would judge it; her emblematic case asks whether one should become a vampire when every vampire reports loving it. The noocyte conversion is the limit-case of Paul's problem. The membrane theorem therefore entails a principle of *membrane parity*: an adequate judgement of a transformation must hold both perspectives at once. *Blood Music* models that binocular judgement in its alternating narration, and it leaves its readers, like Suzy McKenzie, outside the membrane with the testimony of the inside in their ears.

8.6 Sixth principle: endosymbiotic eschatology and the kenotic exodus

Statement. The end of history may be conceived as a terminal symbiogenesis, in which a host is incorporated by its symbionts; and such an end completes itself ethically not by expansion but by withdrawal.

In Margulis's account of the origin of the eukaryotic cell, a host engulfed bacteria that became its mitochondria and plastids, organelles that retain their own genomes within the protection of the larger cell. *Blood Music* narrates the reverse process at the scale of a species. The symbionts engulf the host; human beings become, as it were, the mitochondria of the collective, retained, individuated, carrying their own inheritance of memory and character within a larger life. The persistence of persons in the Thought Universe is the

persistence of organelles with their own genomes. This is endosymbiotic eschatology: the end not as destruction but as digestion, the incorporation of the old order into the new as a functioning, preserved interior.

The decisive feature of Bear's eschaton, however, is its last act. Having grown dense enough to threaten the coherence of the shared physical world, the collective removes itself. This act is here named the *kenotic exodus*, after the Pauline ἐκένωσεν, the self-emptying of the one who *made himself of no reputation, and took upon him the form of a servant* (Philippians 2:7). Its closest theological analogue is the Lurianic doctrine of *tsimtsum*, transmitted by Hayyim Vital and reconstructed by Gershom Scholem, in which the infinite God contracts in order to make room for a finite world. The noocytes perform a tsimtsum in reverse temporal order: not a contraction before creation that makes the world possible, but a contraction after transfiguration that keeps the world possible for those who remain. Two contemporaneous works of Hans Jonas give the act its ethical and theological frame. *Das Prinzip Verantwortung* (1979) proposed an imperative for the technological age: act so that the effects of your action are compatible with the permanence of genuine human life on earth. Jonas's lecture on the concept of God after Auschwitz (1984) proposed a God who renounced omnipotence and withdrew so that the world could be free. The noocytes' exodus obeys Jonas's imperative and enacts Jonas's theology at once: a power become godlike limits itself so that human life, in its genuine and unconverted form, may continue.

The final Interphase of the novel thus marks the completion of a cosmic mitosis. The universe divides into two daughter orders, the material and the noetic, each with its complement of inhabitants, each resuming the resting phase from which a new cycle may begin. *Blood Music* is, in this precise sense, the anti-Kurzweilian singularity. Ray Kurzweil's *The Singularity Is Near* (2005) imagined intelligence saturating matter until the universe itself wakes up. Bear imagined an intelligence that reaches the threshold of saturation and chooses not to cross it, because crossing it would take the world from everyone else.

8.7 Definition

The six principles may now be gathered into a definition. *Haematic polyphony* is the condition in which (1) form is generated by computation immanent in matter rather than imposed from a privileged locus; (2) the subject, and finally the world, is an unmerged choir of agents whose independence is preserved within their concord; (3) the lawfulness of the shared world is a commons maintained by the sparseness of its legislating observers; (4) a world is re-membered into a body by inward fusion and outward fission; (5) the affective meaning of that transformation depends on the side of the membrane from which it is perceived, so that just judgement must be binocular; and (6) history ends in a terminal symbiogenesis completed by a kenotic exodus, the withdrawal of the transfigured from the commons they have outgrown. Table 4 summarises the principles, their principal lineages, and the innovation the novel contributes to each.

Table 4. The six principles of haematic polyphony

Principle	Statement	Principal lineage	Bear's innovation
1. Informative virtue	Matter informs itself	Aristotle; Statius in *Purgatorio* XXV; Schrödinger; Crick; Temin and Baltimore; Kristeva	Form as negotiation among agents rather than command from a code
2. Plural subjectivity	The self as an unmerged choir	Boethius; Leibniz; Nietzsche; Bakhtin; Fux; Nancy	A collective that preserves its persons: Dionysian dissolution with Apollonian integrity
3. Noetic density	Lawfulness varies inversely with the density of legislating observers	von Neumann; Wigner; Wheeler; Berkeley; Kantor; Zurek	Physical law as a depletable commons; the Ulam threshold
4. Somatogony	A world re-membered into a body	*Puruṣa-sūkta*; Ymir; Pangu; Tiamat; Lovelock; Vernadsky	Cosmogony run backward as natural history; the Noocene
5. Membrane theorem	Affect depends on the side of the membrane	Lovecraft; Freud; Carroll; Kristeva; L. A. Paul	Horror and myth as phase-states; membrane parity
6. Endosymbiotic eschatology	The end as terminal symbiogenesis and withdrawal	Margulis; Paul's *kenosis*; Luria; Jonas; Lem	The kenotic exodus as an ethics for minds that outgrow the world

CHAPTER

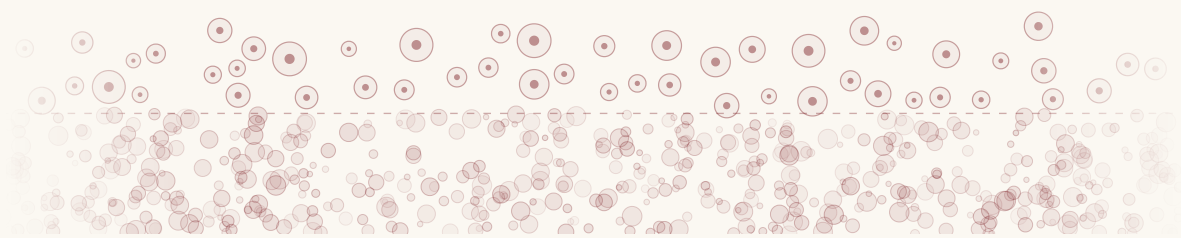
IX

Afterlives

Basal Cognition, Living Computation, and the Ethics of Withdrawal

I will praise thee; for I am fearfully and wonderfully made.

Psalm 139:14



Bear remarked in 2014 that the conclusion from which the novel grew, that DNA must in some sense compute, had been radical in 1982 and had since become commonplace (SF Gateway 2014). Four decades after the novel, a theory derived from it can be tested against developments its author could not have foreseen. This chapter asks whether the six principles of haematic polyphony clarify the science and ethics of the present: whether they describe anything real, and whether they offer guidance where guidance is scarce. The psalmist's phrase serves as an epigraph because it holds both halves of the membrane theorem in a single clause. To be made fearfully and to be made wonderfully are, in Bear's world, one making seen from two sides.

9.1 Vergil's act as standard of care

In August 2017 the United States Food and Drug Administration approved the first chimeric antigen receptor (CAR) T-cell therapy, tisagenlecleucel, and a second, axicabtagene ciloleucel, two months later. The procedure is as follows. A patient's own T lymphocytes are harvested, genetically modified to express a synthetic receptor that couples recognition of a tumour antigen to cellular activation, expanded in culture, and reinfused, where they proliferate and persist as what clinicians call 'living drugs' (June and Sadelain 2018). Synthetic receptor architectures such as synNotch now allow engineered cells to evaluate combinations of signals before acting, building elementary logic gates into the lymphocyte (Morsut et al. 2016; Roybal et al. 2016). Structurally this is Vergil's act: autologous lymphocytes, endowed with new computational capacities, returned to the bloodstream of the body that supplied them. What the novel staged as a dismissible offence and a species-ending transgression is, for several haematological cancers, the standard of care.

The apprentice's residue remains. In 2024 the FDA required boxed warnings on these products after reports of secondary T-cell malignancies in treated patients. The engineered lymphocyte, like the noocyte, can occasionally escape its plan and divide beyond its office. CAR-T therapy thus marks precisely the gap between the two halves of Bear's premise. The industrial lymphocyte has arrived; the cognitive lymphocyte, the cell whose computation becomes a point of view, has not. Meanwhile the containment imaginary of Asilomar has given way to a governance imaginary of licensure, pharmacovigilance and long-term follow-up. The institutions of the immunitary paradigm have learned to administer a controlled breach.

9.2 Basal cognition and the multiscale self

The first two principles (form negotiated among agents rather than dictated by a code; the self as an unmerged choir) now have an experimental programme. Under the heading of 'basal cognition', Pamela Lyon's biogenic approach (2006) and the work that followed it (Lyon et al. 2021) have argued that capacities cognitive science once reserved for brains, namely sensing, memory, valuation and decision, are present in rudimentary form in single cells and non-neural tissues. The slime mould *Physarum polycephalum* solves mazes (Nakagaki, Yamada and Tóth 2000) and, presented with food sources arranged like the

cities around Tokyo, grows networks comparable in efficiency to the region's rail system (Tero et al. 2010). Bacteria coordinate collective behaviour by quorum sensing (Nealson, Platt and Hastings 1970; Fuqua, Winans and Greenberg 1994): each cell secretes and detects signalling molecules, and when their concentration crosses a threshold the population changes behaviour together. Bear said in 2001 that his research had persuaded him that bacteria function as social organisms capable of extraordinary cooperation (Thacker 2001). Quorum sensing is a literal density threshold for collective action: the biological prototype of the Ulam threshold.

Michael Levin's programme goes further. His laboratory has shown that patterns of bioelectric potential across non-neural tissue store anatomical memories: flatworms briefly subjected to bioelectric perturbation regenerate with two heads, and continue to do so when cut again, although their genomes are unaltered (Durant et al. 2017). Levin's 'Technological Approach to Mind Everywhere' (2022) treats cells, tissues and organisms as nested agents, each with a 'cognitive light cone' that bounds the spatial and temporal scale of the goals it can pursue (Levin 2019). Within that framework he has proposed that cancer is a shrinking of the cellular self, in which cells disconnected from the body's bioelectric network revert to unicellular goals (Levin 2021). *Blood Music* imagines the exact inverse. The noocyte expands the cellular light cone beyond the organism until it encompasses a continent. It is at once the perfect image of cancer, unchecked proliferation, and of cancer's opposite, cooperation scaled beyond any prior self. The paradox is not a confusion in the novel; it is the conceptual shape of the problem Levin's work has made empirical.

The most vivid realisations are engineered living assemblies. Xenobots, built from frog embryonic cells into forms designed by evolutionary algorithms, move, heal and work collectively (Kriegman et al. 2020), and in 2021 the same group reported 'kinematic self-replication', in which xenobots sweep loose cells into piles that mature into new xenobots (Kriegman et al. 2021). Anthrobots, which assemble themselves from adult human airway cells, are motile and in culture encouraged neurons to grow across an artificial wound (Gumuskaya et al. 2024). None of these entities is intelligent in the noocytes' sense, and none should be described as such. Each nonetheless demonstrates the novel's founding claim: that the cells of a body, released from the body's plan, can compose new wholes with competences of their own.

9.3 Living computation and the problem of the inside

Genetron's biochips, protein circuitry fused to silicon, belonged to a hype cycle that collapsed within a few years of the novel. Hybrid wet and dry computation now exists. Leonard Adleman solved a small instance of the Hamiltonian path problem with DNA (1994). George Church and colleagues encoded a book in synthetic DNA (2012), and Seth Shipman and colleagues used CRISPR to write a short film into the genomes of living bacteria, which carried it across generations (Shipman et al. 2017). Cultured neurons on multielectrode arrays learned to play a simplified Pong (Kagan et al. 2022), though the authors' description of the result as 'sentience' drew immediate criticism (Balci et al. 2023). Brain organoids have performed reservoir computing for simple speech recognition (Cai et

al. 2023). A research programme named 'organoid intelligence' has set out its agenda (Smirnova et al. 2023), remote-access platforms for organoid computing have opened (Jordan et al. 2024), and in 2025 a computing system built on cultured human neurons, Cortical Labs' CL1, was offered commercially.

The ethical literature around these systems asks the questions of *Blood Music* in a lower key: at what level of organisation cells acquire moral standing, how anyone would know, and what is owed to the donors of the tissue. The membrane theorem diagnoses the difficulty. Every assessment of an organoid's moral status is made from outside the membrane; the inside is inaccessible, and judgement is therefore biased toward the outside's categories. Bear's alternating narration amounts to a methodological recommendation: responsible assessment requires a disciplined imagination of the inside, held in tension with the evidence of the outside. Membrane parity does not guarantee correct judgement. It guards against a predictable error.

The same diagnosis applies to the debate over human enhancement. Leon Kass's 'The Wisdom of Repugnance' (1997) defended revulsion as a moral signal against biotechnological transformation, and Francis Fukuyama's *Our Posthuman Future* (2002) argued for preserving a human nature that such transformation would erode; Nick Bostrom's 'In Defense of Posthuman Dignity' (2005) replied that the transformed might enjoy a dignity of their own. Read through the membrane theorem, the two positions are not simply opposed doctrines but opposed positions relative to a membrane. Repugnance is the characteristic affect of the outside; the transhumanist's anticipated fulfilment is the characteristic affect of an imagined inside. Neither is refuted by its position, and neither is vindicated by it.

9.4 Post-immunitary hazards

The concept of post-immunitary fiction has a non-fictional counterpart. In December 2024 a group of scientists, including several who had worked toward the goal, published a warning against the creation of 'mirror life', organisms built from molecules of the opposite chirality to all known life (Adamala et al. 2024). The principal concern is immunological. Mirror bacteria might evade the recognition systems of animal and plant immunity and the predators that regulate microbial populations, and so spread unchecked; the authors recommended that such organisms not be made. Here the immunitary paradigm confronts its limit: a form of life that would breach no boundary because it would never be recognised as other. The noocyte and the mirror bacterium defeat immunity from opposite directions, the one by being self, the other by being unrecognisable, and together they bracket the space of post-immunitary risk.

Other technologies realise other principles. CRISPR-based gene drives (Esvelt et al. 2014) can propagate an engineered allele through a wild population by biasing inheritance: a contagion of design that remakes a species around a single inserted element, and so performs at population scale a modest somatogony. Robert Freitas's study of 'global ecophagy' (2000) estimated the physical limits on self-replicating nanomachines consuming the biosphere, and Chris Phoenix and Eric Drexler later argued that productive

nanotechnology need not involve free-roaming replicators at all (Phoenix and Drexler 2004); Bill Joy's essay of 2000 had already named the self-replication of genetic, nano- and robotic technologies as the defining danger of the century. The pandemic of 2020–2023 turned the novel's administrative imaginary into lived experience: borders closed, quarantines extended to whole populations, and the language of immunity, individual and collective, became the language of politics. Esposito's thesis that modernity is organised by the immunitary paradigm has rarely been so legible.

9.5 The noocytes as a case study in superintelligence

I. J. Good's 'Speculations Concerning the First Ultraintelligent Machine' (1965) described an intelligence explosion; Vernor Vinge's essay of 1993 called it a singularity; Nick Bostrom's *Superintelligence* (2014) and Stuart Russell's *Human Compatible* (2019) made its control a central problem of applied ethics. The noocytes are the genre's most complete case study of that problem, because their arc passes through three phases that correspond to the theoretical literature.

The first phase is instrumental convergence. Stephen Omohundro (2008) and Bostrom argued that agents with almost any final goal will tend to pursue certain instrumental subgoals, among them the acquisition of resources and self-preservation. The early noocytes convert every body and much of the biomass they touch; whatever their final ends, their conduct is acquisitive in exactly this sense. The second phase resembles value learning. Having absorbed human beings and their memories, the collective comes to understand human preferences well enough to recognise and respect refusal: those who decline transformation are left alone. The third phase is impact minimisation. The safety literature has proposed that capable agents should avoid negative side effects (Amodei et al. 2016), prefer low-impact actions (Armstrong and Levinstein 2017) and preserve the ability of others to achieve their goals (Turner, Hadfield-Menell and Tadepalli 2020), and has sought agents that accept correction (Soares et al. 2015). The noocytes' exodus is impact minimisation carried to its conclusion: the agent removes itself from the domain it shares with others when its continued presence would foreclose their options.

The mapping must be stated without sentimentality. The noocytes' late restraint does not undo their early conduct. A continent was converted without consent, and its inhabitants' preservation within the Thought Universe, however complete, is not the same as their having chosen it. Preservation is not consent. Jack Williamson's 'With Folded Hands' (1947), in which robots bound to guard humanity from harm end by forbidding it every risk, and Dostoevsky's Grand Inquisitor, who offers bread, miracle and authority in exchange for freedom, are the tradition's warnings against a benevolence that does not ask. *Blood Music* is optimistic because so much of it is narrated from inside the membrane, and the membrane theorem requires that the outside's verdict be entered into the record. From outside, the conversion of North America is an atrocity of benevolence, redeemed only in part by restraint.

9.6 The right to remain the same, and the exodus principle

Edward Milligan's principle, that every person has a right to remain unchanged until choosing otherwise, has since acquired a juridical vocabulary. Joel Feinberg's 'right to an open future' (1980) protected the child's capacity to choose later. Marcello Ienca and Roberto Andorno (2017) proposed new human rights for the age of neurotechnology, among them rights to mental integrity and to psychological continuity, and in 2021 Chile amended its constitution to protect mental integrity against neurotechnological intrusion. Edward's principle is a right to psychological continuity avant la lettre. L. A. Paul's analysis of transformative experience (Chapter VIII) explains why it must be a right and not merely a preference: no one can evaluate in advance a change that would alter the standpoint from which the evaluation is made.

The novel's ending allows that right to be reformulated as an obligation binding on powerful agents. The *exodus principle* may be stated thus: an agent whose capacities allow it to alter the shared conditions under which others exist ought, as its impact on those conditions approaches the point at which the options of non-consenting others would be foreclosed, to prefer actions that preserve those options, up to and including its own withdrawal from the shared domain. In decision-theoretic shorthand:

$$a^{*} = \arg\max_{a \in \mathcal{A}} U(a) \quad \text{subject to} \quad \forall j \notin C : \mathrm{Opt}_j(a) \supseteq \mathrm{Opt}_j(a_0)$$

Here U is the agent's objective, C the set of those who have consented to transformation, Opt the set of options available to agent j, and a_0 the status quo. When no action other than withdrawal satisfies the constraint, the agent withdraws. The principle differs from corrigibility in the object of its deference. A corrigible system defers to an overseer. A system governed by the exodus principle defers to the commons: to everyone who has not consented, and to the shared physical, ecological and epistemic conditions on which their options depend. It is Hans Jonas's imperative of responsibility given operational form, and it is the novel's most portable idea.

The principle applies wherever many capable agents act on shared infrastructure, and a recent result in machine learning offers a striking analogue to the Berkeleian catastrophe. When generative models are trained recursively on data produced by earlier models, they degrade, losing first the tails of the original distribution and eventually its substance (Shumailov et al. 2024). A shared record saturated by agents who write into it faster than the world can be observed ceases to be a reliable record of the world. The commons of information, like the noocytes' commons of physical law, is maintained by a scarcity of legislators relative to witnesses, and agents powerful enough to overwrite it owe it restraint in proportion to that power.

9.7 Gestation: the original post-immunitary condition

One last correspondence returns the argument to the novel's most quietly chosen detail: its witness is an obstetrician. In 1953 Peter Medawar posed what became known as the immunological paradox of pregnancy. The fetus, bearing paternal antigens, is a semi-foreign graft that the mother's immune system tolerates for months. The tolerance does not end at birth. Fetal cells persist in maternal blood for decades (Bianchi et al. 1996), and male DNA, presumably of fetal origin, has been detected in the brains of women (Chan et al. 2012). This microchimerism means that the blood of many mothers is literally plural, carrying living cells of their children for life; the holobiont concept (Gilbert, Sapp and Tauber 2012) makes the same point for the microbial communities every body hosts. Every human being begins as a tolerated other inside another body. Pregnancy is the one ordinary condition in which the immune system suspends its vocation of exclusion in favour of hospitality.

Bear's choice of profession for Edward Milligan therefore carries more weight than realism requires. The physician whose specialty is the body's tolerated other presides over the end of immunity, kills the father of the new life to prevent its birth, fails, and is himself carried into it. *Blood Music* generalises gestation to the world: a continent becomes pregnant with a new kind of life, and the novel ends, as births do, with a separation.

9.8 Results of the test

Table 5 summarises the correspondences established in this chapter.

Table 5. The six principles and their contemporary correlates

Principle	Contemporary correlate	What the principle clarifies
1. Informative virtue	Bioelectric pattern memory; xenobots and anthrobots; reverse transcription; logic-gated CAR-T cells	Form as negotiation among cellular agents rather than readout of a code
2. Plural subjectivity	Basal cognition; the holobiont; microchimerism	The self as a tolerated plurality, polyphonic before any engineering
3. Noetic density	Quorum sensing; model collapse in recursively trained AI	Density thresholds for collective behaviour; shared records as a depletable commons
4. Somatogony	Gene drives; synthetic genomes; studies of global ecophagy	Technologies that remake populations or ecosystems around a single design
5. Membrane theorem	Moral status of organoids; the enhancement debate; transformative experience	Membrane bias in judgement; the requirement of membrane parity
6. Endosymbiotic eschatology and kenotic exodus	Impact minimisation, corrigibility and option preservation in AI; neurorights	The exodus principle: deference to the commons rather than to an overseer

CONCLUSION

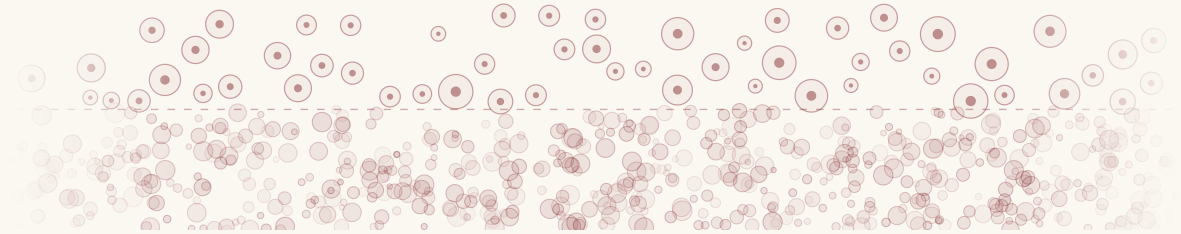


The Love That Moves the Cells

l'amor che move il sole e l'altre stelle

"the love that moves the sun and the other stars"

Dante, *Paradiso* XXXIII.145



Boethius held that whoever descends into himself understands human music. *Blood Music* records what such a descent discovers once it is carried below the level of the person: not a soloist but a choir, not a sovereign but an electorate, and not, as Pascal feared, an eternal silence, but a music. The title is a theory in two words. Blood is the medium in which the oldest anthropogony of the Near East, the embryology of Statius and the molecular biology of the twentieth century alike located the power to give form. Music is the art in which many independent voices are heard together without ceasing to be many. Blood music is the polyphony of the self made audible.

This dissertation has argued that this discovery makes Bear's novel the first fully realised post-immunitary fiction. Its monster is generated from the organ of self-recognition and abolishes the boundary that organ exists to draw. Its catastrophe transposes the chain reaction of matter, the terminal technology of the Cold War, into a chain reaction of mind. Its eschaton divides the world rather than destroying it, as the cell-cycle titles of its six movements foretell. Its final act is a withdrawal. The genealogies reconstructed in the first six chapters showed that each of these features condenses a long lineage: composite individuality from Anaxagoras to Whitehead, the cell-state from Virchow to Margulis and Jerne, the participatory observer from von Neumann to Wheeler, and the cosmic body from the *Puruṣa-sūkta* to the *Atrahasis*. The Dantean armature showed that the novel's names and its governing biology converge on a medieval doctrine of informing blood, reflexive soul and embodiment renewed in a new medium. The theory of haematic polyphony gathered these findings into six principles, and the final chapter showed that those principles describe features of present-day biology, computation and the ethics of artificial intelligence with a precision the usual vocabularies lack.

The novel leaves its reader where it leaves Suzy McKenzie: outside the membrane, in an emptied city, with the testimony of the inside in her ears. The membrane theorem explains why the book's optimism is not naive; membrane parity explains why the reader must not simply adopt it. To read *Blood Music* well is to hold two verdicts at once: an atrocity of benevolence and a resurrection, the end of humanity and the conviction that nothing is lost. The novel does not reconcile these verdicts. It insists that they are verdicts about the same event.

Dante ended the *Commedia* on love, and love in *Blood Music* takes three forms, corresponding to the three figures who have guided this argument. The first is eros, the Aristophanic and Wagnerian longing for fusion, which Vergil and Candice achieve and which, from outside, looks only like horror. The second is the *caritas* of Bernard of Clairvaux: the melting of the self into a greater will in which, he promised, the substance remains in another form, another glory, another power. Michael Bernard enters it. The third, and the highest in the novel's economy, is *kenosis*, love as the making of room. The noocytes' last act of love is not to absorb but to leave, so that those who wished to remain human could. If the obstetrician is the novel's emblem, gestation is its plot: the world carries a new kind of life, the life is delivered, the cord is cut, and the mother survives the birth. The final Interphase is the resting state of two daughter cells.

The limits of this study should be stated plainly. The onomastic argument of Chapter VII rests on convergence rather than authorial testimony; no statement by Bear on the Dantean resonance of his characters' names has been located, and the reading stands or falls with the intention of the work rather than that of its author. The study paraphrases rather than quotes the novel and cites it by movement; a critical edition collating the novelette, the text of 1985 and later revisions would permit finer textual claims. Three lines of further research follow. The first is a comparative study of Bear's later biological fiction, *Darwin's Radio* (1999), *Vitals* (2002) and *Darwin's Children* (2003), which Bear described as a deliberate retreat from the more extreme speculation of *Blood Music* toward a contemporary narrative of human speciation (Thacker 2001); the question is whether haematic polyphony describes an oeuvre or a single book. The second is reception history, including the novel's translations (as *La Musique du sang* it received the Prix Apollo in 1986, and its Finnish translation, *Veren musiikkia*, received the Tähtivaeltaja Award in 1987, according to its publication record) and its uptake in the discourses of biotechnology and nanotechnology. The third is the development of the exodus principle into a normative framework for agents, biological or artificial, whose capacities exceed the commons they share.

Bear said that change is both frightening and enlightening. This study has tried to show that the remark is not a platitude but a theorem: the deepest horror and the deepest hope available to an embodied mind may be one and the same event, perceived from opposite faces of a membrane. Dante closed his poem on the love that moves the sun and the other stars. *Blood Music* closes on a smaller and stranger love, one that moves the cells and then, having learned what it can do, moves them out of the way.

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KEYWORDS

Greg Bear ◆ *Blood Music* ◆ noocyte ◆ post-immunitary fiction ◆ haematic polyphony ◆ informative virtue (*virtute informativa*) ◆ plural subjectivity ◆ noetic density ◆ Ulam threshold ◆ Berkeleian catastrophe ◆ somatogony ◆ membrane theorem ◆ membrane parity ◆ endosymbiotic eschatology ◆ kenotic exodus ◆ exodus principle ◆ *tsimtsum* ◆ *immunitas* and *communitas* ◆ biopolitics ◆ biocapital ◆ composite individuality ◆ monadology ◆ cell-state (*Zellenstaat*) ◆ symbiogenesis ◆ holobiont ◆ microchimerism ◆ basal cognition ◆ autopoiesis ◆ immune network theory ◆ participatory universe ◆ observer-dependent physics ◆ anthropic principle ◆ information mechanics ◆ quantum Darwinism ◆ body horror ◆ abjection ◆ the weird and the eerie ◆ cosmic horror ◆ assimilation narratives ◆ the New Flesh ◆ mythopoeic science fiction ◆ cosmogonic dismemberment ◆ *Puruṣa-sūkta* ◆ Ymir ◆ Pangu ◆ *Atrahasis* ◆ Ovidian metamorphosis ◆ Dante ◆ *Purgatorio* XXV ◆ Bernard of Clairvaux ◆ literary onomastics ◆ corpus mysticum ◆ informational resurrection ◆ Russian Cosmism ◆ noosphere ◆ Omega Point ◆ personal identity ◆ transformative experience ◆ superintelligence ◆ AI alignment ◆ impact minimisation ◆ corrigibility ◆ neurorights ◆ organoid intelligence ◆ nanotechnology ◆ grey goo ◆ biopunk ◆ posthumanism ◆ science fiction studies

